

ReCiPSS

D2.3-Circular business models evaluation reports

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List of abbreviations

<i>Abbreviation</i>	<i>Explanation</i>
BM	Business model
BMI	Business model innovation
BMD	Business model development
CBM	Circular business model
CBMD	Circular business model development
CBMI	Circular business model innovation
FMEA	Failure Mode and Effect Analysis
IT	Information technology
IoT	Internet of thing
IS	Information system

1. Executive Summary

This deliverable presents reports from the evaluation of creation, delivery, and capture of the circular value at different stages of developing circular business models (CBM) of the demonstrators: Bosch/C-ECO and Gorenje. The evaluation process maps initial assumptions, plans, provided analysis results, and necessary organizational changes realized during the previous development phases. Due to the complex character of the process of circular business model innovation and given that the innovation was a long-term project, the mix of methods and techniques has been chosen.

For the evaluation first the Methodology had to be defined. The aim of the Methodology is to capture and evaluate management aspects of the development and implementation of circular business models and to identify barriers and challenges of this process. Methodology reflects basic knowledge of several areas related to the management of the project preparation process, the development of circular business models and their gradual implementation in the business life of demonstrators. For the Methodology the literature review was conducted. The aforementioned complexity also affected the content and scope of review, which included following areas: existing approaches to the evaluation of business model development from the point of view of project management; basic characteristics of change management; selected aspects of innovation management (precisely the requirements of the individual phases of managing business model innovation) and risk management and risk assessment as an integral part of project management and business model development. Specific features of circular business models and their development that should be acknowledged during the whole process from the initiation to the implementation of the development/innovation of circular business models have been also taken into the consideration in the Methodology. The methodology developed in this way has made it possible to capture the breadth and depth both of the process and project of the circular business models development and implementation of the demonstrators. The Methodology is also used to create Guidelines presented in the final part.

For the evaluation itself inquiring as the method with a mix of two techniques was employed. Qualitative approach in the form of in-depth interview with the three representatives of Bosch and quantitative approach in the form of survey with the respondents from Gorenje and C-ECO took into account the level of involvement and the level of innovation of business models of the demonstrators.

In the final part, the deliverable contains the Guidelines for systematic risk assessment in the implementation process of circular product systems. Guidelines are developed based on the detailed documented Methodology used in the evaluation, results of the evaluation and existing knowledge from the literature review. Guidelines reflect the existing deviations between the assumptions, plans and the real processes, and results coming from the evaluation.

2. Introduction

This document is a result of Task 2.4: *Evaluating the circular business models of the demonstrators* that focuses on the evaluation of the creation, delivery and capture of circular value at different stages of the demonstrator progress, taking into account all previous development stages, assumptions, plans, analysis results and necessary organizational changes. The second task is to develop Guidelines for systematic risk assessment in the implementation process of circular product systems based on the Methodology proposed and used in the evaluation process.

Figure 1: Linkages between Task 2.4 and other tasks and working packages shows how this task is linked to other tasks in the WP2 and to tasks in other packages.

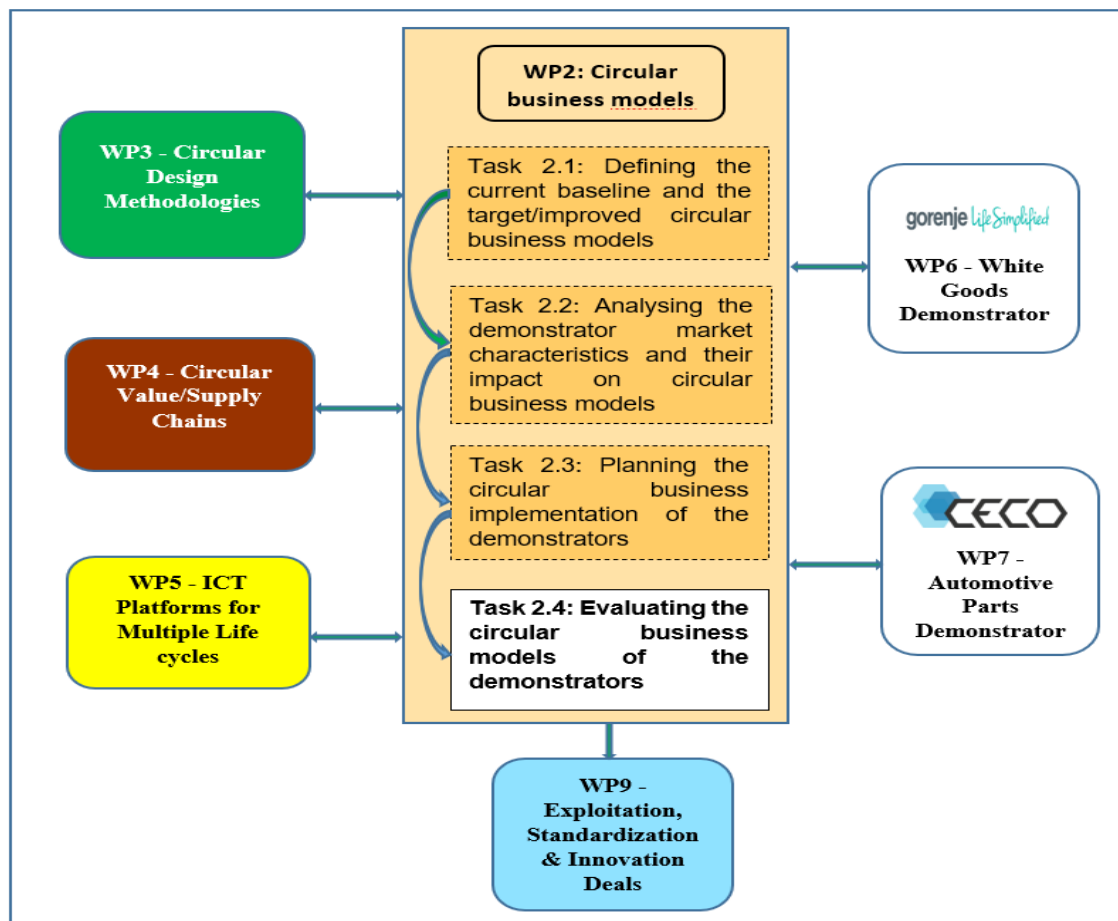


Figure 1: Linkages between Task 2.4 and other tasks and working packages

Task 2.4 builds on the results from all previous tasks in WP2 and some outcomes from the realized tasks from other work packages:

- WP 3: Circular Design Methodologies (in particular from existing results of co-creation workshops with consumers, core-brokers and remanufacturers)
- WP4: Circular Value/Supply Chains (outcomes from Task 4.1 and 4.2 related to the managerial aspects of circular supply chains and networks' operations)

- WP5: ICT Platform for Multiple Life cycle (information from Task 5.3 and 5.4 about the ICT platform integration into the processes of both demonstrators were an important input into the evaluation)
- WP6: White Goods Demonstrator
- WP7: Automotive Parts Demonstrator (particularly, numerous discussions and intensive collaboration with partners have provided insight into specific issues in existing processes that may be relevant to the management of circular business model development and thus relevant for the evaluation task).

For Task 2.4, knowledge of circular business models and innovative tools from the systematic literature review (Task 2.1), characteristics of demonstrators' markets and their potential impact on the circular business model (Task 2.2) and issues of the circular business model implementation (Task 2.3) were relevant for understanding the challenges in the business model development and implementation process.

Outputs from Task 2.4 should become inputs to Task 3.5: Circular design implementation and evaluation report – for the part dealing with the analysis of the role of organizational structure and changes necessary to support the transition to circular systems. Further to Task 4.4: Training key stakeholders and evaluating the performance of the circular value/supply chains, where the evaluation of factors concerning the circular value delivery within the Task 2.4 may bring an important insight. Both Task 6.7 and 7.5: Evaluating the technical, socio-economic and environmental performances of the demonstrator may involve summarising the results from the interviewing demonstrators to evaluate circular business models and process of development and Innovation. Key information from Deliverable 2.3 may also be inspiring for Task 9.2: Developing business cases and upscaling plans – for the overall evaluation of the circular business models 'performance at the end of the project.

2.1 Methodology

For the task, several methods and techniques were employed. The first one is a brief literature review of the following areas:

1. *Project management and project evaluation* – the reason for providing a review in this area was the project character of ReCiPSS and the related need to incorporate aspects of project management and project evaluation.
2. *Organizational change and innovation and business model innovation management* – this review brings insights into the main characteristic features and challenges of the approaches and their linkages, a summary of the innovation processes and justification of the approach applied in this deliverable.
3. *Risks management and risks assessment* – this part of the review provides the necessary overview of the knowledge needed to elaborate the Guidelines of the systematic risks assessment.
4. *Evaluation of the circular business model development and innovation* – this review aimed to summarize specific features of circular business models, which should be taken into consideration when defining the evaluation criteria as well as specific barriers, challenges and risks for the development and innovation of circular business model, which were also included into the evaluation.

The second one is interviewing demonstrators with the combination of survey (Gorenje and CECO) and personal interviews (all three companies – together with Bosch - as demonstrators). For the analysis of results from the evaluation (interviewing), the gap analyses were performed based on the aggregation of the individual responses to the questions from questionnaires. Personal interviews were analyzed in the form of a qualitative content analysis.

For the guidelines of the systematic risk assessment in the implementation process of circular product systems, the aforementioned literature review and results from the interviewing and gap analyses were used.

Figure 2 presents the research framework¹.

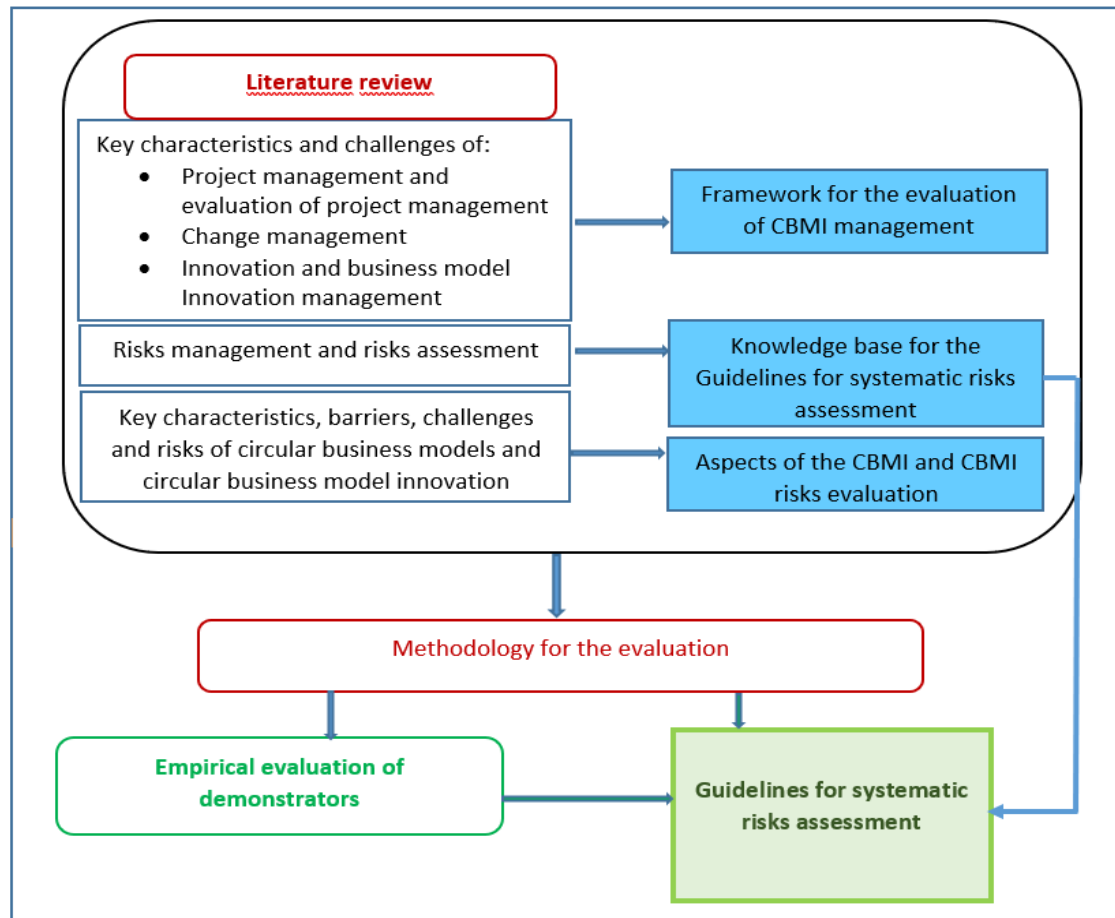


Figure 2: The research framework

2.2 Document Scope

This deliverable consists of several parts. The first two (Chapter 3 and Chapter 4) are dedicated to the brief literature review on evaluating projects in the context of change and innovation management, risk management and assessment, and of specificities of evaluating the circular business models development and innovation. Review is needed to:

¹ Acronym CBMI in the Figure means circular business model innovation

- a) contextualize proposed Methodology and Guidelines for the systematic risks assessment;
- b) to summarize key aspects of managerial and organizational challenges (barriers and risks) of any and specifically circular business model development and innovation; and
- c) to create an overview of key areas for the evaluation of past and current processes and situation and for the guidelines for the systematic risk assessment of the implementation process of circular product systems.

Detailed Methodology for the development and innovation mapping process follows in Chapter 5. Next, the results from mapping (interviewing) are presented and discussed in Chapter 6. The last part (Chapter 7) contains the Guidelines for the systematic risks assessment in the implementation process of the circular product systems as the circular business models. In Conclusions, we discuss some limitations of the proposed Guidelines for the systematic risks assessment and Methodology applied for the evaluation and finally, we formulate some ideas for future development.

3. Circular business model evaluation and risks assessment

3.1. Project evaluation - overview of the basic principles and approaches

Companies can design their circular business models from scratch. They can also acquire existing business model, but established companies more often adjust, innovate, redesign, transform or diversify their existing models (Gauthier and Gilomen, 2016; Santa-Maria et al, 2020). Such companies sometimes operate two or more business models at once when one or more of them could be developed as a very new, more or less complex model (Guyader and Piscicelli, 2019). This means that circularity can be a radical versus an incremental innovation of the current business. Only some business model components can be adapted versus the whole one; novelty may relate more to the internal aspects of the business model, or on the contrary, novelty can apply to the entire external network (Santa-Maria et al, 2020). For the following text, the terms business model development (BMD) and innovation (BMI) are used most often for the expression of different approaches the companies apply when turning their attention towards circular economy challenges in business and for business models (Tesch et al, 2017).

According to Bocken et al (2014) the key to business model innovation is the quality of management and leadership. Incorporating such human-behaviour aspects in terms of integrating knowledge and skills of project management, organizational change, and innovation management is one of the crucial approaches to understanding the How and Why questions of the success of any business model development or innovation (Pieroni et al, 2019). Many of the business model development and/or innovation takes the form of project management with clear start and end time, with a more or less specific team of people and more or less clear tasks, processes and activities, authorities and responsibilities, allocated and usually multiple resources across the organizational lines, schedule with planned phases, performance requirement, element of risk and established goals (Munns and Bjeirmi, 1996; Smith and Imbrie, 2004).

Projects are the means of implementing strategies (Kenny, 2003) and, as such, are always accompanied by changes. Conscious change management is considered one of the pillars of successful project management (Hornstein, 2015). Moreira et al (2016) believe that change is a partner of innovation. The authors work on the commonalities or shared features and differences between both concepts. They argue that (p. 136) “while for change the defining focus is on the parameters altered by the change, for innovation, the focus shifts to the qualitative results obtained”. A close bond between project management and innovation is stressed among all by Davies (2013).

3.2. Project management and project evaluation

Project evaluation is one of the pivotal parts of project management (Heyworth, 2002), playing an essential role in **organizational learning and performance measurement** (Ingason and Jónasson, 2009), and in **exercising control** (Laursen et al, 2017). It is the assessment and analysis of a project's effectiveness by comparing the project's effects at a given point in time. Evaluation can take place:

- a) *ex-ante* - when usually criteria for the evaluation and assumptions for the comparison and selection of the alternative scenarios and/or implementation are formulated;
- b) *interim (or ongoing or mid-term)*, which main aim is to intervene in the ongoing project if necessary; and
- c) *ex-post* (post-implementation) with the highest level of the learning effect – in short, to take lessons and insights) against the project's objective (Van der Veen, 2010; Linzalone and Schiuma, 2015).

Ex-post and interim evaluation can also be referred to as *prescriptive evaluations*. Both are conducted to **assess** the project: **the impact of what has been implemented** and lessons for further process improvement. Their purpose is to provide a comparison of planned with actual achievements to learn how well resources have been used, whether the initial assumptions about what would happen were correct, and to improve the management of subsequent phases or in the case of another project (Thomas et al, 2007; Van der Veen, 2010). Literature shows that post-implementation evaluations are rarely carried out because they are either perceived as a formality after the project is ready or because of missing knowledge and methods of how to do it or lack of time and interest (Thomas et al, 2007; Al-Yaseen et al, 2008) despite the value they might bring. The other barriers concern a lack of data because - during the project - not all needed data used to be collected and a barrier related to knowledge management or non-existence of **knowledge management**. Especially for larger and longer project duration, people do not remember the past. The problem of the nonlinear nature of the knowledge generation process, the difficulty of extracting tacit knowledge – if it was remembered and realized at all - and the difficulty of identifying relevant knowledge for the past is more challenging with the time passing (Christiansen and Mouritsen, 2021).

The evaluation can help improve ongoing (or future) projects or project management; evaluation itself aids managers to keep track of their projects and project performance (Laursen et al, 2017). Evaluation for the purpose of the improvement enables understanding **what and how has been achieved** (Linzalone and Schiuma, 2015) and **what not** and if possible, why not. Evaluation should also answer **the questions where and why the problems or gaps exist within project life**.

From the methodological point of view, project evaluation could also be understood as “a set of research questions and methods properly articulated to review processes, activities and strategies” (Linzalone and Schiuma, 2015, p. 92). Evaluation applies processes of measuring, analysing, and interpreting change.

Evaluation should reflect and also incorporate the knowledge of reasons **why project and project management fail**. There are several reasons for the failures (Altuğ, 2002, p. 24; Jenner, 2015, pp. 4-5; Todorovič et al, 2015):

- problems with organizing project team and overall governance;
- weak project leadership;
- communication problems and insufficient exchange and flow of information and knowledge sharing;
- conflict and resolution;
- insufficient upper management involvement;
- difficulties in defining work in sufficient detail;
- lack of a clear link between the project and the organization's key strategic priorities, including agreed measures of success;

- lack of effective engagement with relevant stakeholders who participate or are affected by the project or the results of the project;
- lack of skills and proven approach to project management and risk management;
- no previous experience and lessons learned or inadequate use of the existing ones;
- insufficient knowledge management;
- too little attention to breaking development and implementation into manageable steps;
- preference of short-term financial benefit and outcome before the long-term benefits;
- the separation of strategy formulation ('choosing') from implementation ('doing');
- adopting activity-based rather than results or benefits-led change;
- failing to address transition management adequately and to include all required business changes;
- the project built on the (unsafe) foundations of unrealistic forecasts.

The evaluation **criteria must be defined**. Borrowing from the OECD document (2019, p. 5), criteria are used in evaluation to: 1. "support accountability, including the provision of information" to the relevant stakeholders; to 2. "support learning, through generating and feeding back findings and lessons", and 3. "also beyond evaluation – for monitoring and results management, and for strategic planning and intervention design. Criteria can be used to look at processes (how change happens) as well as results (what changed)".

Haas and Guzman (2020) point out that existing project evaluation literature neglects soft criteria (that reflect "temporal, dynamic, complex and subjective nature of today's project", which are more complex than before) and is concentrated on objective, which is relatively easy to measure, and more on tangible aspects of projects. Literature also does not cover evaluation of project understood as socially constructed phenomena. As the projects are developed and realized by people, arguments articulated by Christiansen and Mouritsen (2021) are noteworthy. In their understanding, a project is an organizational (and many times also inter-organizational) experience and, as such, the outcome of multiple, complex construction processes of different people who are directly involved in the process or who are affected by the project. In the evaluation, different and often also competing narratives should be reflected, and the effort to understand may lead to the improvements of project management in the future.

Most approaches consider the so-called golden triangle in project management (time, cost and quality of the project) and do not work with criteria such as impact on society and the environment. The latter issue is directly linked to the ReCiPSS project as those impacts are an integral part of the project and must be included in the evaluation.

Based on the systematic literature review of project evaluation criteria, Haas and Guzman (2020) summarized four categories of criteria: effectiveness and efficiency, business success, impact, sustainability. Christiansen and Mouritsen (2021) identified 38 post-project evaluation criteria, organized into four groups – regarding objectives, regarding internal processes, regarding knowledge management and finally regarding outcome evaluation. Even though evaluation for this deliverable has an interim character, some of these criteria and the logic can be applied for the reports.

Evaluation as a process has several features and phases. The answers to the basic questions (why, what, who and for whom, how and when or about the goals, content and scope,

beneficiaries and evaluators, time and method) should be obtained. In more details, this means following tasks and phases (Constanzo, 1975; Galas et al, 2018):

1. Clarification of why the evaluation is to be carried out, what is the purpose, and what are the objectives of the evaluation (long-term and short-term benefits of a project);
2. Identifying what is to be evaluated;
3. Identifying what information is needed to conduct the evaluations;
4. Defining who should evaluate and who should be evaluated;
5. Defining the criteria and measures for the evaluation;
6. Selecting or developing the instruments to collect the necessary information;
7. Defining when the evaluation should be done; and
8. Developing a management system of the evaluation process.

3.3. Organizational change and innovation management

Organizational change management is one of the key elements of the circular business model innovation process (Santa-Maria et al, 2021). Change management can be defined as „*the process of continually renewing an organization's direction, structure, and capabilities to serve the ever-changing needs of external and internal customers*“ (Moran and Brightman, 2000, p.66). The process of organizational change is very complex and challenging (Whelan-Berry and Somerville, 2010).

According to Anderson and Anderson (2010), three critical focus areas need to be integrated for a successful change – content, people, and process. **Content** indicates what is about to change, e. g. strategy, structure, systems, processes, technology, products and services, work practices, as well as whole business models. As Jäderberg and Kottorp (2019) explain, the components of any business model are highly interdependent; thus, changing one will probably result in changing others as well. **People** refer to the humans involved in the change process – those who are designing, implementing, supporting, or simply being impacted by the change. This aspect includes their thoughts, emotions, behaviours, motivation, culture, communication, relationships, etc. (Anderson and Anderson, 2010). Oreg et al (2011) consider the reaction of change recipients (people) to be the primary determinant of the success of the organizational change. However, Chiappetta Jabbour et al. (2019) argue that this „human side“ is underrepresented in the research concerning the circular economy. The last focus area suggested by Anderson and Anderson (2010) is the **process**, denoting decisions and actions that produce the content and people outcomes.

From a **project management perspective**, a general change management process consists of **five phases**. The process starts with **identifying the need for a change**. During this phase, the current, the future and the transition state are decided. In the second phase, the **change details regarding how activities will be done, and roles will be changed are determined, and cost and risk analyses are performed**. The next phase involves assessing stakeholders' needs and interests and creating action, communication, **and resistance plans for the change management process**. The fourth stage consists of implementing the plans, which is when the transition state occurs. And finally, during the **monitoring phase, the changes are controlled**, making sure they are leading up to the desired state. **Errors and lessons learned are recorded for the future** (Galli, 2018). Burke (2017) points out that in the planning phase, the change process usually looks linear. However, despite the effort to follow the planned steps, the

implementation phase is often anything but linear due to the occurrence of unanticipated consequences. The author stresses the need to have clearly defined goals and change leaders willing to stay in a course that may help the process to become somewhat linear eventually.

Change management models

In addition to the general change management process inspired by project management described above, numerous models focused on implementing best the planned organizational change (Stouten et al, 2018). Kurt Lewin, widely considered the father of change management, came up with a model consisting of three steps – unfreeze, change and refreeze (Cummings et al, 2016). For the implementation, a disadvantage is that the model does not provide details on dealing with the human aspect of the change (Galli, 2018).

Beer et al (1990) propose a critical path – a sequence of **six steps to an effective change**: 1. *mobilize commitment to change through joint diagnosis of business problems*, 2. *develop a shared vision of how to organize and manage for competitiveness*, 3. *foster consensus for the new vision, competence to enact it, and cohesion to move it along*, 4. *spread revitalization to all departments without pushing it to form the top*, 5. *institutionalize revitalization through formal policies, systems and structures*, 6. *monitor and adjust strategies in response to problems in the revitalization process*. The authors emphasize that it is important to respect the order of steps since it might be counterproductive if a step starts too early. The unique feature of this model compared to others is the attention brought to monitoring and adjusting the change (Stouten et al, 2018).

Kotter's (1995) model includes **eight steps**: 1. *establishing a sense of urgency* to motivate a large number of people to cooperate, 2. *forming a powerful guiding coalition* – this coalition should also include members who are not part of senior management, 3. *creating the vision* – a picture of the future appealing to stakeholders and relatively easy to communicate, 4. **communicating the vision** – the author suggests that „*communication comes in both words and deeds and the latter is often the most powerful form*“ (p. 64), 5. **empowering others to act on the vision** which includes communication but also removing obstacles to the new vision, 6. *planning for and creating short-term wins*, 7. *consolidating improvements and producing still more change*, 8. *institutionalizing new approaches*. The author further suggests that change is stable when anchored in corporate culture.

The **ADKAR model** proposed by Hiatt is a framework for understanding change at an individual level (Hiatt, 2006). Stouten et al (2018) point out that it focuses on the process that helps employees become ambassadors of the change, taking into account their individual needs. Hiatt (2006) explains that the model comprises **five building blocks** that all must be in place for a successful realization of a change. *Awareness* means that an individual understands the nature of the change, the reason for the change being made, and the risk of not realizing the change. *Desire* refers to the willingness to support the change and engage in it. *Knowledge* stands for the information, education and training needed to know how to change. *Ability* means turning knowledge into action, i.e., realizing the change. *Reinforcement* represents factors sustaining the change. They might be external such as rewards or recognition, and internal in the form of satisfaction or other personal benefits. Galli (2018) believes that because the model focuses mainly on the human side of the change, it is more appropriate for project teams than for large organizations with complex processes.

McKinsey 7S framework, introduced by Peters and Waterman, depicts a constellation of seven interrelated factors influencing an organization's ability to change (McKinsey&Company, 2008). The model comprises „hard“ factors – strategy, structure and systems, and „soft“ factors – shared values, skills, staff and styles (Ravanfar, 2015). All the factors are considered to be on the

same hierarchical level, which suggests that a change in one of them will require working on the others (McKinsey&Company, 2008).

Resistance and Readiness

Armenakis et al. (2007) state that many change failures are a result of employee resistance. According to Self and Schraeder (2009), some authors even consider it the most important threat to the successful implementation of strategic initiatives. Resistance can be viewed as a multidimensional comprising behavioural dimension (how people behave in response to change), cognitive dimension (what they think about the change) and affective dimension (how they feel about the change) (Erwin and Garman, 2010). Piderit (2000) suggests that employee's reaction to change may differ across dimensions. For instance, beliefs about a change might be more positive than the emotional response to it.

Galli (2018) proposed that not all resistance to change must be wrong since it forces management to review their vision or roadmap to identify problem areas. Similarly, Ford and Ford (2009, in Simoes and Esposito, 2012) argue that *“resistance, properly understood as feedback, can be an important resource in improving the quality and clarity of the objectives and strategies at the heart of a change proposal. And, properly used, it can enhance the prospects for successful implementation”*.

Armenakis and Harris (2009) propose that rather than waiting to reduce resistance, organizations should focus on creating readiness for change which they define as *„the cognitive precursor of the behaviours of resistance to or support for organizational change“* (Armenakis et al., 1993, in Armenakis and Harris, 2009, p. 132). Weiner (2009) argues that organizational readiness for change is a multi-level construct that might exist at individual and group levels. In the latter case, it can be viewed as *„a shared psychological state in which organizational members feel committed to implementing an organizational change and confident in their collective abilities to do so“* Weiner (2009, p. 6). Armenakis (1999, in Armenakis et al., 2007) suggested five key change sentiments necessary to support change readiness, adoption and institutionalization. *Discrepancy* stands for the belief that there is a significant gap between the current and the desired state (Armenakis and Harris, 2009); therefore, a change agent should take actions to illustrate the need for change (Armenakis et al. 2007). This is in line with Kotter's (1995) proposition that a change effort should start with establishing a sense of urgency. *Appropriateness* refers to the belief that the proposed change will effectively address the discrepancy. *Efficacy* means the confidence in individual and organizational abilities to succeed in implementing the change (Armenakis et al., 2007). *Principal support* refers to the belief that formal leaders and opinion leaders are committed to the success of a change. *Personal valence* relates to the perceived benefits of the change for the recipient (Armenakis and Harris, 2009).

Innovation and business model innovation management

Within the ReCiPSS project, the demonstrators are in a different situation concerning the level or the intensity of their circular business model innovation. While Gorenje develops a complex new circular business model, which is substantially different from the existing model Gorenje currently has, C-ECO innovates only some features of their operations and implements a new tool (ICT cloud-based platform) for the cores evidence and traceability. However, the new circular value will be offered to current and new customers, processes of value delivery will be adapted, and new value capture will be solved. This means that in the case of Gorenje, many new areas in the internal and external environment have to be managed and organized for the project's success. In contrast, in the case of the other demonstrator, the impact and needed interventions are smaller, and they only partially affect the existing business model. However,

incremental innovation also requires managers' attention, especially if there are some new activities in the processes or some changes within some business model elements.

Business model innovation represents a novel and more holistic form of organizational innovation (Foss and Saebi, 2017); it is meant to be a type of **organizational change** (Huang et al, 2013). Common characteristic with the organizational change is the **organizational inertia and resistance**, which negatively affects innovation of business model and adaption to the new way of conducting business because of the **disruption of the organization's current interests and power structure** (Huang et al, 2013).

There are several characteristics of business model innovation that distinguish this type of innovation from the others (product, process, organizational, market or management). First, it is crucial for managers to understand the **complexity of the interplay between innovation and the elements of BMI** (Hossain, 2017) **and between BMI and strategy** (Brink and Holmén, 2009). Second, innovation of business model affects an organization on a broader scale because it – in many cases - requires **organizational restructuring** (Hossain, 2017). Third, the involvement of **top management** is imperative in this type of innovation as well as the role of **leadership**. Fourth, **BMI is also more complex** as it often needs **different types of innovation** to be managed (innovation of products, processes, markets, organizational aspects and management) (Massa and Tucci, 2013). Fifth, it is also a dynamic system that requires flexibility and continually adjusting to internal and external changes. Sixth, with its scope, BMI affects and requires **cooperation from more departments and more functions and people across the organization** than product or process innovation. Seventh, it is also more open innovation because the need for inputs and **cooperation with the external stakeholders** is higher. Eighth, it is also riskier, **more resources and capabilities (dynamic capabilities) and time** demanding compared to another type of innovation (Brink and Holmén, 2009; Hossain, 2017; Teece, 2018). Ninth, unlike any other innovation type, especially product innovation, **every BMI is unique**, and **most of the needed routines do not exist** (Björkdahl and Holmén, 2013). Tenth, the question of **who is responsible for business model innovation is rather complicated**, although the answer should be clear – top management (Chesbrough, 2007). Eleventh, while innovation in the form of products and services can often easily be copied, “business model innovations allow companies to change the rules of the game” (Bucherer et al, 2012). Twelfth, while a test and testing are concepts associated with product innovation and many processes and market or organizational innovations, for BMI the concept of experiment and experimentation is more appropriate (Chesbrough, 2010; Zott and Amit, 2015). McGrath (2010 in Schneckenberg et al, 2017, p. 406) described it as „a discovery process that undergoes phases of Marketplace experimentation and tentative resource allocation to maintain firms’ decision making capability“. Twelfth, business model innovation may be triggered but also hindered by **organizational culture**. Hock et al (2016, p. 433) found out that “novelty-oriented cultural values foster capabilities (strategic sensitivity, collective commitment and resource fluidity) in favor of business model innovation, while efficiency-oriented cultural values do not show positive effects.”

Business model innovation is a process (Zott and Amit, 2015; Foss and Saebi, 2017), and it must be managed so that it results in a successful new or innovative model. Hartley (2006, in Frankenberger et al, 2013, p. 253) justified the importance of knowing the character of the sub-phases of the business model Innovation process as follows: *“the articulation of processes helps to identify particular barriers and facilitators at particular stages, and this may be of practical help to policy-makers and managers.”*

The literature review reveals many approaches to BMI as a process that vary in number and content or focus of phases, sequence, and a set of performed activities (Tesch et al, 2017). For

instance, Wirtz and Daiser (2018) identified 20 different business model innovation processes. They show that while some processes are rather BMI design-oriented, other processes focus on the operations or management and realization of BMI. Pieroni et al (2019, p. 201) identified 92 approaches, and they categorized them into three broad stages: “(1) sensing: identifying opportunities and generating new BM ideas; (2) seizing: systematically designing and testing new BM concepts or configurations; and (3) transforming: building new competences and implementing organizational renewal.” In 2021 the authors added one process before sensing: “prepare” (Pieroni et al, 2021). For circular economy and business model innovation recently published British standard BS 8001:2017 *Framework for implementing the circular economy principles in organizations. The guide* is inspiring. The BMI comprises 8 phases: Framing, Scoping, Idea generation, Feasibility, Business Case, Piloting and prototyping, Delivery and implementation, and Monitor, review and report. The process could be a sequential one, but it is recommended to understand it as the “plan-do-check-act” cycle, emphasizing the possibility and need of repeated stages during the whole innovation process. (Paliuk, 2018, Chen et al, 2020).

For the evaluation of business model innovation, the processes and activities before the idea generation (one of the most often accepted start to innovation process) should be included as they reflect the strategic management behavior (one characteristic feature of BMI) and capabilities of managers. This is also – to some extent - the case of BS 8001:2017 and the first two phases of Framing and Scoping. However, for the evaluation process to go through 8 phases is rather demanding.

Frankenberger et al. (2013) suggested a slightly different and simplified set of BMI phases, called 4I Framework, based on process models from the innovation management literature and insights from business model case studies. The 4 phases identified are **initiation** (understanding the ecosystem – “the discovery of the need for innovation. That is, the capturing of the initial event, idea, or decision that initiates the entire innovation process”, p. 256); **ideation** (generating new ideas – “how to react to the impulse”.....“aims at opening up the solution space and at generating a set of possible alternatives“, p. 256); **integration** (aligning the business model internally – “takes up one of the promising possibilities and focuses on its elaboration and development”....to turn the (selected) idea into some tangible product, process, or service”“the idea is embedded into and integrated with a broader context.“, p. 256), and **implementation** (making investments – and “brought to the market“, p. 256). In discussing these phases, the authors focused more on the **challenges** than on the activities performed by business model designers. (Zott and Amit, 2015). The phases and the Framework are built on the logic of a feedback loops, and there may be iterations between the phases. The 4 phases are open and can be “filled” with different sub-processes and activities of relevance for business model innovation as needed (Björkdahl and Holmén, 2013). 4I Framework will also be applied in the Methodology for the demonstrators' circular business model development and innovation.

Typical activities within the **Initiation** phase are market analyses to sense and discover opportunities and acknowledge challenges, threats, and risks and to understand drivers and barriers for the strategic focus and directions for the company. Opportunities identified in the previous phase are in the **Ideation** phase transformed into the alternative relatively concrete business models. This means that opportunities are matched with the existing resources and capabilities and/or the potentialities of acquiring the new resources and capabilities. In the **Integration** phase, the promising ideas identified in the ideation phase need to be transformed into a complete and viable business model. Specifically, the revenue model (‘Why’) and value chain architecture, i.e., resources, partners, activities (‘How’), must be solved in this phase as the added elements of What (value proposition) and Who (customer) thought out in the

previous phase. It is vital to convincing relevant stakeholders to support new business models, become committed, and adjust their business models accordingly. Active management of partners is needed. Besides that, adapting existing and new supply chains and networks, adapting existing and acquired new resources, and redesigning or designing new activities must be orchestrated. The main task of the **Implementation** phase is to convince people, make them understand how and why the new business model can be beneficial, working, and profitable, and not be afraid of the associated changes and trial-and-error process of this phase (Frankenberger et al., 2013).

Following Table 1 summarizes challenges of the individual phases of business model innovation identified by Frankenberger et al (2013).

	Business model innovation phases			
	Initiation	Ideation	Integration	Implementation
challenges	Understanding of the needs, pain points of the players and opportunities in the company's ecosystem	Difficulty in overcoming the current business logic and in thinking out-of-the-box	Integration of all pieces of new business model	Overcoming internal resistance
	Identification and mastering of change drivers	Difficulties to think in business models No systematic tools to develop new business model ideas	Involvement and management of partners	Management of the chosen implementation approach

Table 1: Challenges of business model innovation phases

3.4. Circular business model innovation and risk management and assessment

Circular business model development and innovation to become circular one is both associated with many general and specific drivers, perceived and expected benefits (see for instance Linder and Williander, 2017; Tura et al, 2019; Gusmerotti et al, 2019; Bocken and Gerds, 2000), but also with many challenges, barriers, and risks that accompany the whole process of development and innovation. Hofman and Jaeger-Erben (2020) claim that “transitions to CBMs are highly uncertain projects akin to a black box”. Knowledge of those challenges, barriers, and risks is crucial for managers to incorporate strategic and operational thinking and acting into the developing and innovating activities and avoid as many potential problems when implementing new or innovated business model as possible.

In practice it is sometimes difficult to make clear distinction between challenges, barriers or hindrances and risks because all three previously mentioned areas could be perceived and considered as potential risks. However, according to some authors risk as opposed to the uncertainty is possible to be identified, predicted, and calculated or evaluated in advance (Reim et al, 2016). These subprocesses are parts of risk management as will be discussed below.

In a project, the “risk is a measure of the degree of possible variation in the outcome or benefits of the project” (Teltumbde, 2000). Risk in more general understanding refers to the “likelihood of events of a negative nature” (Brillinger et al, 2020) and can be defined as an „uncertain future event which could influence the achievement of the organisation's objectives, including

strategies, operational, financial and compliance objectives“ (Jallow et al, 2007, p. 168). Coming back to the uncertainty, for instance Hopkin (2018) defines risk as the „uncertainty of outcomes“, definition typical for control risks, i.e. risks associated with project. Risk is also understood as positive or negative outcome of uncertainty (Olsson, 2007) or an “effect of uncertainty on objectives” with both positive and negative consequences of uncertainty (ISO 31:2018).

Any business model is accompanied by risks of a different nature. Brillinger et al (2017, p. 9) characterize business model risks as „all risks within the business model which can endanger the profitability and sustainability of the business model or even company goals and value“. This definition refers to the risks of the business model as such, but risks may also be present in individual elements of business model and affect other elements to a greater or lesser extent (Shi and Manning, 2009).

Managing risk is an integral part of right corporate management and decision-making at all levels of an organization. What is different is the degree of systematicity, formality, focus, details, consistency and if it is managed continuously or randomly (Berg, 2010). “Risk management is a systematic approach to setting the best course of action under uncertainty by identifying, assessing, understanding, acting on and communicating risk issues” (Berg, 2010, p. 81). It should be seen as an important tool for managers to make the most appropriate decisions at different hierarchical levels, taking into account the factors, forces and specificities of the internal and external organizational environment (Rampini et al, 2019). According to the ISO 31000:2018 Guide “The risk management process involves the systematic application of policies, procedures and practices to the activities of communicating and consulting, establishing the context and assessing, treating, monitoring, reviewing, recording and reporting risk.”

Risk management can be divided into sub-phases to make it easier to grasp management activities and assign roles needed to ensure risk management. There is no agreement which phases are included in risk management. For instance, Cagliano et al (2015) for project risk management identify six phases (planning, identification, analysis, response, and monitoring and control). The Project Management Institute’s Guide to the Project Management Body of Knowledge (PMBOK Guide, 6th version, 2017) defines project risk management processes as: plan risk management, identify risks, perform qualitative risk analysis, perform quantitative risk analysis, plan risk responses, implement risk responses and monitor risks. ISO 31000:2018 Guide contains six phases starting with establishing the scope, the internal and external context and criteria for risk management; risk assessment (which includes risk identification, risk analysis and risk evaluation); risk treatment; risk recording and reporting; risk monitoring and review and risk communication and consultation. According to Boehm (1991) risk management (in general) consists of two main phases: risk assessment, which includes sub-phases of identification, analysis and prioritization, and risk control, which includes sub-phases of risk management planning, risk resolution and risk monitoring planning, tracking and corrective action. We combine his approach with the approach of Altuğ (2002, p. 9) for the project management who identified several general processes and added one sub-phases of risk assessment (evaluation) from ISO 31000:2018 Guide. Combination is introduced in Table 2 below. According to Becker (2004) a feedback loop exists for the process. For the final phases ISO 31000:2018 Guidelines suggest several options how to treat risk: avoiding the risk by deciding not to start or continue with the activity that gives rise to the risk; accepting or increasing the risk in order to pursue an opportunity; removing the risk source; changing the likelihood; changing the consequences; sharing the risk with another party or parties (including contracts and risk financing) and retaining the risk by informed decision. However, prior to the decision making to choose the appropriate way, risk assessment must be done.

Steps			Statement
Risk assessment	Risk Management Planning		deciding how to approach and plan the risk management activities for a project
	Risk identification		determining which risks might affect the project and documenting their characteristics
	Risk analysis	Qualitative risk analysis	performing a qualitative analysis of risks and conditions to prioritize their effects on project objectives
		Quantitative risk analysis	measuring the probability and consequences of risks and estimating their implications for project objectives
	Risk evaluation		comparing the results of the risk analysis with the established risk criteria to determine where additional action is required
Risk response development			developing procedures and techniques to enhance opportunities and reduce threats to the project's objectives
Risk monitoring and tracking, control, and corrective actions			monitoring residual risks, tracking the emergence on new unforeseen risks, and identifying them, executing risk reduction plans, evaluating their effectiveness throughout the project life cycle and provide corrective actions

Table 2: Model of a project risk management processes

Processes of risk assessment will be borrowed and adapted for the purpose of the systematic risk assessment of circular business model development and innovation management.

Collins dictionary (2021) defines risk assessment as the evaluation of possible risks in a product, situation, activity or course of action. The primary objective of risk assessment is to estimate risk by identifying an event (mostly undesired), the likelihood of occurrence of the event, and its consequence or impact (KarimiAzari et al, 2011). This process can help to select less risky projects, avoid, or minimize risky steps and eliminate the residual risk (Barghi and Sikari, 2020). As Aven (2016, p. 4) argues “The way we understand and describe risk strongly influences the way risk is analyzed and hence it may have serious implications for risk management and decision-making.”

First step of risk assessment is **risk management planning**. Within this step some areas should be clarified as how the risk will be handled, responsibilities within risk management, how, what and when plans will be developed, realized, controlled, and adapted, if needed. Zwikaël and Ahn (2011) argue that in project management planning is a critical phase and a critical tool for uncertainty reduction and that most risk management practices should be performed during the planning phase of a project. The authors also present several drawbacks of risk management planning, specifically the existence of limited variety of tools used (with dominating risk events ranking) when in case of new and more complex project existing tools need not to be sufficient. Second, poor performance of risk planning processes concerning risk identification and development of effective strategies is rather frequent in project management.

To be able to manage risk, **risk identification** must be done as the second step using some of the risk identification tools. According to ISO (in Aven, 2011) the risk identification is the “process of finding, recognizing and describing risk”, and involves the identification of risk sources, events, their causes and their potential consequences. For a networked environment, as it is in case of CBM, Halikas et al (2004) argue that risk identification must take into account the dependencies on other organizations. Altuğ (2002) divides tools into two major categories: experience-based and brainstorming-based while for instance Marle and Gidel (2012) into the analogical, heuristic, and analytical approaches. More specifically, the tools include, but are not limited to checklists, scales, diagramming techniques (e.g. cause-and effect diagrams, affinity diagrams), documentation reviews (if existing), PESTLE and SWOT analysis brainstorming, interviewing experience or relevant persons, Delphi expert judgement etc. Major threats, challenges, barriers, and opportunities that can affect the project processes and outcomes are detected and described (Barghi and Sikari, 2020). For business model development and innovation because of its complexity a categorization or classification of risks is appropriate across whole model and within and between elements of business model (Brillinger, 2018). Classification could be the approach applied for the identification and could serve also as a tool after the identification to group risks for easier management and task allocation. Identification of risks in the initiation and particularly in the ideation phase of BMI is associated with the **identification of sources** of uncertainty leading to potential risks, while reconsidering the existence and character of risk and identification of new risk emerging due to changes in the environment is typical for the phases of implementation and integration of business model (Cagliano et al, 2015).

Risks can be classified in different ways, for instance into internal and external risk, into the strategic, operational and cultural risk (Taran et al, 2015), to market, country and also global risk, to compliance and environmental risk (Toma and Alexa, 2012) or into following five areas (cf. Půček et al, 2021): (1) financial (related to the uncertainties with cash flow, ability to invest, ability to meet financial commitments and include for instance budgeting, tax matters, accounting, investment, pricing etc. are managed); (2) organizational and personnel (related to, for example, management and leadership level of competence and experience and errors, competence or lack of staff, poor timing of activities, incorrect deadlines, etc.); (3) legal and regulation (especially current or forthcoming legislation, internal regulations); (4) technical and technological (for example, existence and quality and capacity of IT and ICT infrastructure, machines, equipment and tools for remanufacturing or refurbishing); (5) material (relating to the sustainability of spare parts, the condition, quality and availability of inputs to the circular business model operations etc.). Kovaitė and Stankevičienė (2019) also added competence and data security risks. Some other risks (e. g. procurement, delay, safety, design physical and force majeure risks) are introduced in Stosic et al (2017). Among the external and market risks, often of strategic nature competitive customer and reputation risks belong (Nordin et al, 2011). Reim et al (2016) deal besides technical risks also with behavioural (both internal and external) and delivery competence risks. Strategic risks may affect company’s sustainable development, growth trajectory, shareholder value and success of BMI. Operational risks are related to the reduced ability to produce and supply products and services (Nordin et al, 2011). Another approach of risks categorization is presented by Epstein and Rejc (2006, p. 8). Different categories and types of risk are summarized by these authors into the broad category of the organizational risks (see Figure 3).

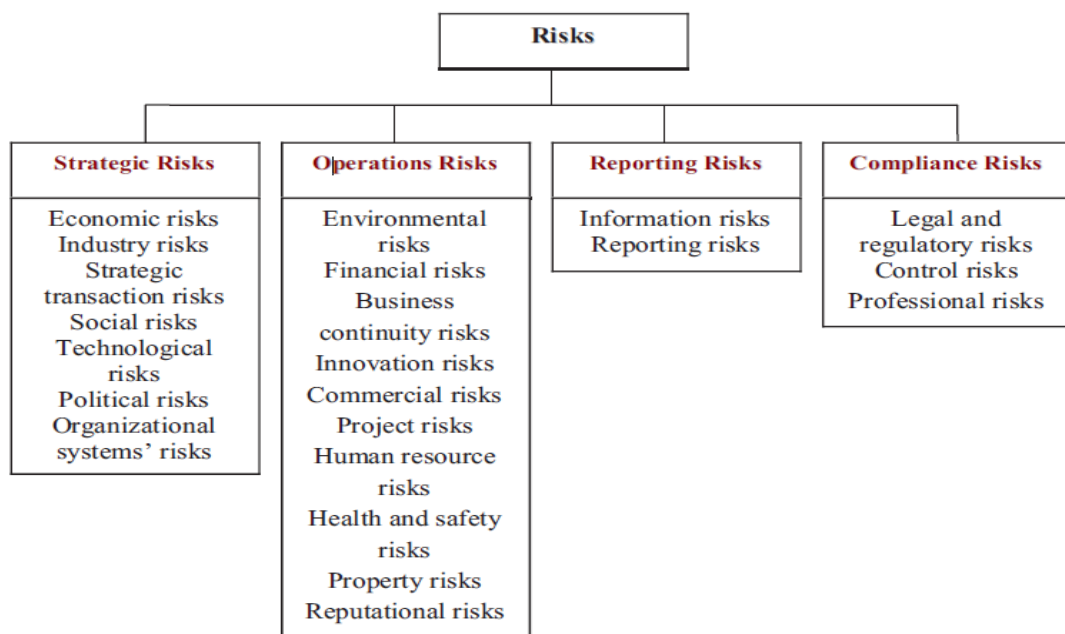


Figure 3: Risks categorization

The result of the risks identification is a list of risks divided into the areas. The identification of risks should be documented, and document(s) should be managed. Risks and nature of risks are not unchanging and permanent. Just on the contrary there are new risks emerging during the process of business model innovation as well there are risks that no longer exist or there is changed. Systematic risk assessment is a constantly evolving process and persons responsible for the business model innovation (or project) must continuously assess the existence of risks.

Examples of how the risk can be categorized and documented are in following three tables.

Category 1	No.	Description of risk
Financial risks	1.1	Lack of investment funds
	1.2	Increase of operating costs
	1.3	Accounting errors
	1.4	Exchange rate threats
	1.5	Rising input prices

Table 3: Examples of identified financial risks

Category 2	No.	Description of risk
Organizational (management) and human risks	2.1	Staff shortages
	2.2	Insufficient or non-existent experience
	2.3	Inappropriate or non-existent incentives
	2.4	No or low commitment of top managers
	2.5	Poorly defined goals, poorly delegated tasks, ...

Table 4: Examples of identified organizational and personnel risks

Category 3	No.	Description of risks
Legal and regulation risks	3.1	Non-compliance with applicable laws and regulations, specifically
	3.2	Non-existing standardisation, standards, legislation
	3.3	Significant differences in legislation in different countries
	3.4	Lengthy approval of legislation
	3.3	Contracts

Table 5: Examples of identified legal risks

After identifying the main risks, the third step involves the accurate **assessment of risks**. There are different approaches how to assess risks from the very simple to the very complex and software demanded approaches. The frequency of the occurrence, impact, probability of occurrence of risks are most typical variables coming from FMEA method and its variants (e. g. Carbon and Tippet, 2004; Marle and Gidel, 2014). However, for the business model innovation frequency is probably less appropriate.

A qualitative assessment of the risk helps to evaluate the risk verbally – within the context and including various dimensions of assessment. Brainstorming, scenario analyses, Delphi technique, cognitive maps, ranking, matrixes (Parnell, 2009) are the most common methods and techniques for this type of assessment. The above mentioned three variables are used here. Impact is usually more concretely specified as impact on...schedule, quality, relationships etc. (Mojtahedi et al, 2010).

On the contrary, especially if many and diverse types of risks exist or will potentially exist, quantitative risk assessment is applied. This requires some level of calculation, and it is useful for prioritizing different risks (Halikas et al, 2004). Probability and decision trees, probability risk assessment, MOda technique, Monte Carlo (Parnell, 2009), MAGDM technique (Mojtahedi et al, 2010) are the most frequent methods, yet require some mathematical and statistical knowledge.

Examples of qualitative assessment involving rating scales are shown in Table 6 and Table 7 (Halikas et al, 2004, p. 53).

rank	estimate	description
1	No impact	Insignificant in terms of the whole company
2	Minor impact	Single small losses
3	Medium impact	Causes short-term difficulties
4	Serious impact	Causes long-term difficulties
5	Catastrophic impact	Discontinue business

Table 6: Impact assessment scale

rank	estimate	description
1	Very unlikely	Very rare event
2	Improbable	There is indirect evidence of event
3	Moderate	There is direct evidence of event
4	Probable	There is strong direct evidence of event
5	Very probable	Event recurs frequently

Table 7: Probability assessment scale

For the quantitative risk analysis and for the risk evaluation Strategic Issue Management (SIM) matrix suggested by Ansoff (1980) could be applied. It is rather simple tool originally developed for strategic change management and combining two perspectives: impact and urgency. Impact means the estimated consequence of the risk, which has negative (or positive) character and relates to the objective of the issue. “Urgency is a qualitative measure of the time left to resolve an issue” (Ansoff, 1980). It is based on the estimated difference in time between when an issue’s impact interferes, and in the time that is needed to resolve the issue (Malinen, 2020). A Matrix is shown in Table 8.

	IMPACT		
URGENCY	low	significant	major
low	drop from the list	Periodic review	Monitor continuously
significant	Drop or periodic review	Periodic review or monitoring	Plan delayed response
pressing	monitor	Monitor continuously	Respond immediately

Table 8: Assessment of the expected urgency and impact of risk (urgency/impact matrix)

Final step is risk evaluation which means to compare risk management plan with the outputs of risk identification and assessment.

3.5. Evaluation of the circular business model development and innovation

Circular business model development and innovation to become circular one is both associated with many general and specific drivers, perceived, and expected benefits (see for instance Linder and Williander, 2017; Tura et al, 2019; Gusmerotti et al, 2019; Bocken and Gerdt, 2000), but also with many challenges, barriers, and risks that accompany the whole process of development and innovation. Hofman and Jaeger-Erben (2020) claim that “transitions to CBMs are highly uncertain projects akin to a black box”. Knowledge of those challenges, barriers, and risks is crucial for managers to incorporate strategic and operational thinking and acting into the developing and innovating activities and avoid as many potential problems when implementing new or innovated business model as possible.

However, for a CBM, as Linder and Williander (2017) argue, it is more difficult to proactively evaluate critical assumptions because of the length of the time horizon of the impacts model implementation. It is generally much more difficult to formulate all potential critical assumptions because of a higher level of uncertainty, higher level of complexity in case of CBM and missing knowledge of various or even many circumstances because there are not many existing real-life examples from practice yet.

According to Santa-Maria et al (2021) development and innovation of CBM is a complex innovation challenge when companies move in unknown terrain. It is considered to be also an organizational change process (as with the traditional BMI), where changes – in case of more radical innovation or a new model development - occur across the whole organization, reconfigure existing supply chains and networks or even lead to the development of totally new supply chains and networks. This feature is more pronounced with circular as with the “traditional” business model innovation.

The specific challenge for the CBMI is the requirement of the capability to measure and evaluate progress on circularity performance in different contexts and levels (Franco et al, 2021). This means, for instance, that circularity performance should be approved not only in the internal environment of a company but also at the supply chain and partners’ level and – if possible – also at the macro level. The company is no longer focused on profitability, but profitability comes from circularity and other social and environmental benefits.

As risks also barriers can be evaluated. Guldmann and Huulgaard (2020, p. 10) present a comprehensive review of existing circular business model innovation barriers (see [Table 9](#)).

Internal barriers	External barriers
The unclear business case for CBMs Integrating environmental considerations in product and business model design is also perceived as lengthening time to market Concerns over the risk of decreasing sales due to increased sales of repaired, reconditioned and remanufactured products Product design should follow certain guidelines to enable circularity, and redesign of old products may thus be needed	Lack of concrete, coherent, strict legislation No government support in the form of training, funding, legislation. No clear place to go for help and long procedures to obtain certifications etc. Lack of supportive public procurement policies Taxation of labour rather than raw materials renders labour-intensive reuse and recycling activities expensive

Lack of top management commitment Lack of in-house knowledge about the circular economy, including remanufacturing, recycling etc. and its benefits Not possessing the necessary technical and technological know-how to engage in new business models. The difficulty of designing new business models, including a lack of tools for this Companies' supply chain position may limit opportunities to adopt CBMs. Required expertise and knowledge about the products make CBMs most suitable for OEMs Lack of internal resources, i.e. capital, time and staff to investigate opportunities A fundamental shift in corporate culture, policies and market engagement is needed and demand internal reorganisation. Resistance to change Traditional incentive structures and performance metrics are inappropriate to support new business models Repairs impaired by proprietary product designs, parts glued together and other physical product attributes Products and buildings are complex and not designed with end-of-life reuse or recycling in mind resulting in a low value at end-of-life	Legislation hinder CBMs, e.g. legislation on sales of waste materials and on the cross-border movement of products for reuse Warranty legislation hinders the use of reused spare parts Lack of external funding opportunities Financial, legal and operational risk increase in CBMs compared to linear business models, and tools to assess and manage risks are lacking Lack of interest and understanding from the value chain and a need for training and education Network collaboration challenges, e.g. difficulty of creating the needed networks for circularity and supply chain dependencies that prevent circularity Original spare parts are difficult or impossible to attain or have to be transported over long distances The rate of technological change may demand frequent design changes that hinder product reuse and remanufacturing Fragmented supply chains, lack of green suppliers and long distances to customers Unpredictable or decreased quality of returned or recycled products and materials Unpredictable flow of reused goods or recycled materials resulting in, e.g. conflict with demand and planning difficulty. Lack of information systems to mediate this No official training available to repair staff, no access to repair tools and repair guidelines for third-party repair companies. General lack of knowledge experts on circular economy CBMs only relevant to some customers and product types and customer benefits from, and acceptance of, new CBMs are uncertain The low status of products from recycled materials and repaired, reused, refurbished or
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	remanufactured products. Uncertainty about the residual value of products. The low status of products may damage company image if it engages in CBMs Changing fashion trends can be a challenge for long-life products
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Table 9: Internal and external barriers to CBMI

Another risks, challenges, barriers but also critical areas of CBMI will be presented in following text. Overview of them may enable innovative companies to better work with risks.

4. Methodology for the evaluation

Galas et al (2018), based on Stufflebeam (2001), present several evaluation types for research projects, which could be applied for any project and any evaluation, i. e. also for the CBMI evaluation because of the generality of the approach. For this Deliverable, we apply the logic of a combination of two broad approaches and two types of evaluation:

1. **questions-oriented approach** and an *objective-based type of evaluation* that typically focuses on whether the “objectives have been achieved through an internal perspective (by project executors” - in the ReCiPSS project relevant persons of the demonstrators). For the evaluation of the circular business model development, we formulated several questions that are related to the managerial and organizational issues of project life in case of circular value creation, delivery and capture². Limitations of the question-oriented and objective-based type approach consist of the narrow scope to constitute a sufficiently broad and complex basis for evaluating the project life as it concentrates mainly on achieving the objectives.

2. **improvement/accountability-oriented evaluation approach** - here, the reasoning of the *decision/accountability-oriented type of evaluation* was used. Such evaluation aims to help to improve the project (proactively in case of the project on progress or for the learning effect) and/or to assess the merit and worth of the project (retroactively) (Stufflebeam, 2001). The questions addressed within this evaluation were derived from the project goals and the known and assumed stakeholders’ concerns and the literature review on project, change, innovation, and risks management and circular business model development and innovation. Assessment of needs, problems, challenges, barriers, risks and intended and unintended outcomes are typical.

This approach typically involves multiple qualitative and quantitative assessment methods to provide crosschecks on findings and engage stakeholders in the evaluation (Stufflebeam, 2001). As Stufflebeam argues, by involving stakeholders in the evaluation process, the change process can be improved because stakeholders are decision-makers, participants, users of the outcomes in the project. Unfortunately, we could not engage stakeholders from the external environment, which is one of the evaluation limitations. In addition to both demonstrators, we also interviewed three managers from Bosch who are involved in demonstrating the ICT platform in remanufacturing plant in Göttingen. This interview was very short and represented only a fragment of the whole evaluation.

For the evaluation, we used survey (C-ECO and Gorenje) and interviews (Bosch, CECO and Gorenje). The focus of interest is on the management of the project life and related management processes. The complexity of the project and the involvement of more people from one organization lead to the decision to use multiple informants to increase the reliability and validity of the reports (Kumar et al, 1993) and obtain a more complex picture (Walker and Enticott, 2004). Delphi technique approach was involved. This technique requires knowledgeable and expert contributors individually responding to questions (in form of a survey), assess given event, situation, problem, or risk from their experienced perspective. There is rich academic discussion about the methodology of this technique without no clear standardized outcome (Hasson and Keeney, 2011; Sobaih et al, 2012). Delphi technique is appropriate particularly in cases if heterogeneity of participants with different expertise and experience is needed to gain more complex understanding of the researched issue to assure validity (Grisham, 2009) and also to obtain consensus on final judgement or evaluation of the

² See Appendixes A, B and C

topic. The experts are able to focus on rating, revising, and commenting on the items presented in the questionnaire without the distractions normally associated with face-to-face interviewing or focus groups (Latif et al, 2017). Consensus is obtained based on revision of experts' judgement in the further round which could be also in the form of interviews (Sobaih et al, 2012).

For the evaluation of circular business model innovation, we use the evaluation of answers after first round and interviewing to focus on major risks and challenges members of demonstrators' companies perceive with CBMI. To analyse answers from multiple informants in one organization, we follow the recommendations of Kumar et al (1993) and Van Bruggen et al (2002) for the methodology and the aggregation of responses. Aggregating experts' evaluations assess the value of a project based on multiple criteria. (Bohanec et al, 1995). Most of the informants are very intensively involved in the whole project – from the initiation phase, some from the ideation or integration phase.

For the survey, we developed a questionnaire mapping **the managerial and organizational aspects** of the development process of the circular business model. In the questionnaire - for assessing the project evolution and business model innovation process - we also applied the 4I framework (Frankenberger et al, 2003) for understanding phases of the circular business model innovation/development process. For this Deliverable, we also borrow ideas and adapted the circular business model innovation model from the recent comprehensive review of CBMI provided by Santa-Maria et al (2021) (see Figure 4 below). Elements and content of this conceptual model are involved into the questions for the questionnaire and interviews as they enable to meet the requirement of complexity of CBM and CBMI. Circular value creation, delivery and capture are incorporated in the form of specific questions within the 4I framework in the questionnaire, also using elements of the business model canvas. More specifically we map development process also through the lenses of value creation, value delivery and value capture and within the customer value, customer segments, customer channels, customer relationships, key resources, key activities, key partners, costs, and revenues.

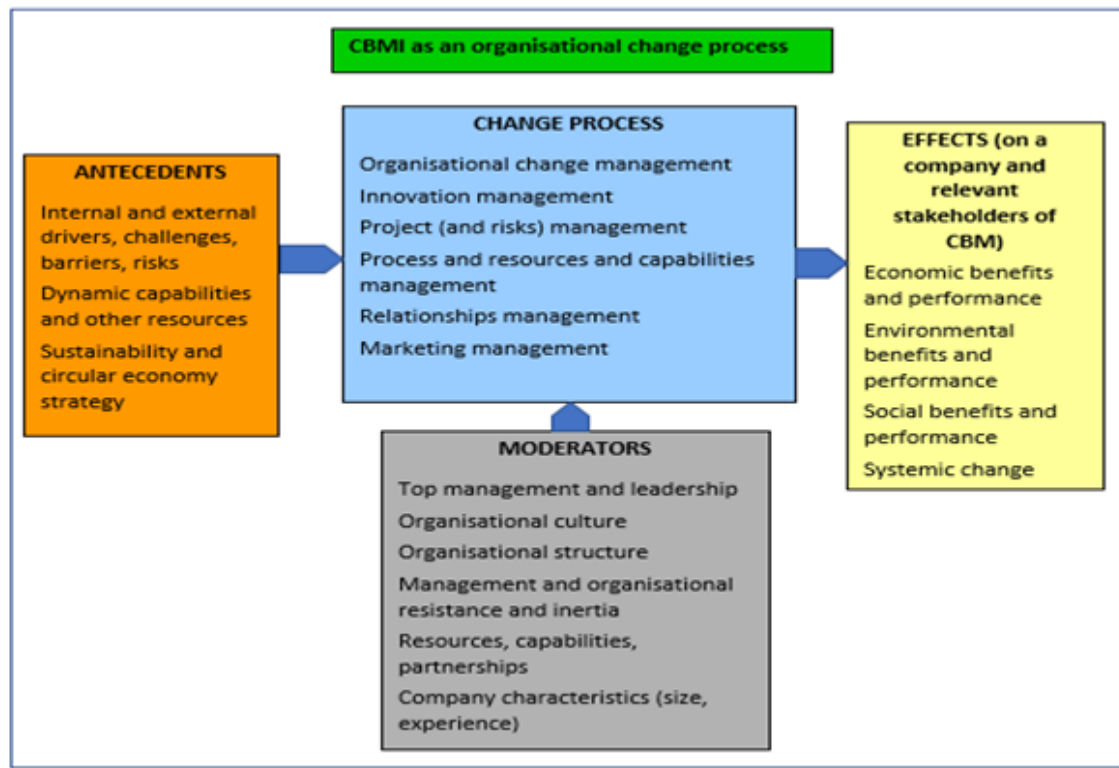


Figure 4: CBMI as an organizational change process

There are two versions of the questionnaire: one for Gorenje and one for C-ECO (see Appendix B and C). The reason lies in the higher degree of radicalness of the innovation of CBM in the case of Gorenje when the whole new circular business model with all elements has to be developed. Therefore, in every innovation process within 4I Framework we also asked about every element of the circular business model canvas. For C-ECO we adapted several questions for the ICT portal, and questions about the individual elements of CBM canvas were raised only in two of four phases of innovation.

Questions were formulated based on the project objectives, literature review presented in this deliverable and knowledge of the team members (WP2) of the project execution during the past 36 months. Both closed, semi-opened and opened questions were formulated. Closed questions serve for the maturity assessment and the development of checklists for the Guidelines for systematic risk assessment.

Closed types of questions involve answers offering measurement of the attitudes, rate of agreement and assessment with four to five-point scales or yes/no answers.

The evaluation objectives are set in a broader context, with inspiration found in the work of Björkdahl and Holmén (2013), who formulated the following questions adapted for our purpose:

- How do companies work with circular business model innovation, and do they do it systematically using existing methods and techniques?
- How are circular business model innovation processes organized, and how well do they function, and are they changed?
- Should companies overcome organizational resistance to innovate their business models?
- Are there, and if yes, what kind of uncertainties and lack of knowledge involved in circular business model innovation?

- Are there any risks perceived with circular business model innovation? Are the risks evaluated (before the project started and during implementation?)
- What are the main challenges and barriers of circular business model innovation?
- How can companies work with new circular business models when they have existing business models up and running?
- What are the main changes during the process of circular business model innovation?

Methodology for the analyses of the answers

The analysis follows the proposed methodology, which - at its very heart - combines the structure of the business model canvas with the 4I-framework of business model innovation (initiation, ideation, integration, and implementation). Both models are explained in the above literature review in detail.

As mentioned above, aggregation of responses, calculation of the averages and analyses of critical areas of circular business model innovation and development were applied for the analyses.

5. Results of the interviewing

5.1. Evaluation results for washing machine demonstration

The evaluation results are presented primarily according to the four phases of the project, i.e., initiation, ideation, integration, and implementation (see the Methodology of the work). It should be taken into account that, in reality, the different phases more or less overlapped and did not follow each other in a purely sequential manner. In each of the phases, both the ratings on the scales (i.e., quantitative) and the relevant verbal statements of the respondents are captured, pointing out some critical or complicated decisions - and for different reasons (typically, however, associated with data and information uncertainty). The rating on the scales is also used in several cases to handle comparisons (gaps) between responses within a given phase. In the case of initiation and ideation, the text also captures the comparisons (gaps) in selected areas between these phases. The aim is thus to highlight differences in challenging aspects of these phases.

Respondents commented on the different Business Model Canvas elements, especially for the ideation, initiation, and implementation phases. Another area included in this analysis is the changes made during the project and the risks management and assessment.

Informants

As people answered the questions with a very close relationship to the project, their responses are synthesized using an arithmetic average rather than a weighted average in the case of the scales. This is because, by virtue of their job position and involvement in the project, some respondents cannot a priori be given more weight than others. The presented outcomes reflect views of three respondents (employees of Gorenje).

5.1.1. Initiation phase

The chance to generate additional revenue was the initial motivation for developing the circular business model (answer "4" on a positive 4 level scale). All that was partially supplemented by the motive of gaining environmental savings (i.e., to produce more durable washing machines). The company wanted to "capitalize on the new business trend". Definitely, the issue of cost reduction played only a marginal role in this (answer "1.5" on a positive 4-level scale).

Understanding the value offered

Within the initiation phase, the most important decisions concerned customer value and circular value proposition. The following table documents that the demonstrator perceived the initial idea of the value, which the new business model should deliver, as relatively clear - a rating of 4 on a scale of 1 to 5 (where 5 meant extremely "unambiguous and clear"). Only in the case of the environmental and social aspects, this idea was less clear. A similar point could be made for the perception of demand, which was not unambiguous, too. However, this was not the case for the detailed understanding of market segments, which was not based on real data but rather on "stereotypes and trends in other industries".

There were two major challenges associated with this phase; the first was to get sufficient support from people in the company because the new business model meant a shift from a pure manufacturing company towards a service-oriented one. The second challenge concerned the technical development of the washing machines (connected devices) and the entire IT infrastructure (supporting the new business model, including the ordering system).

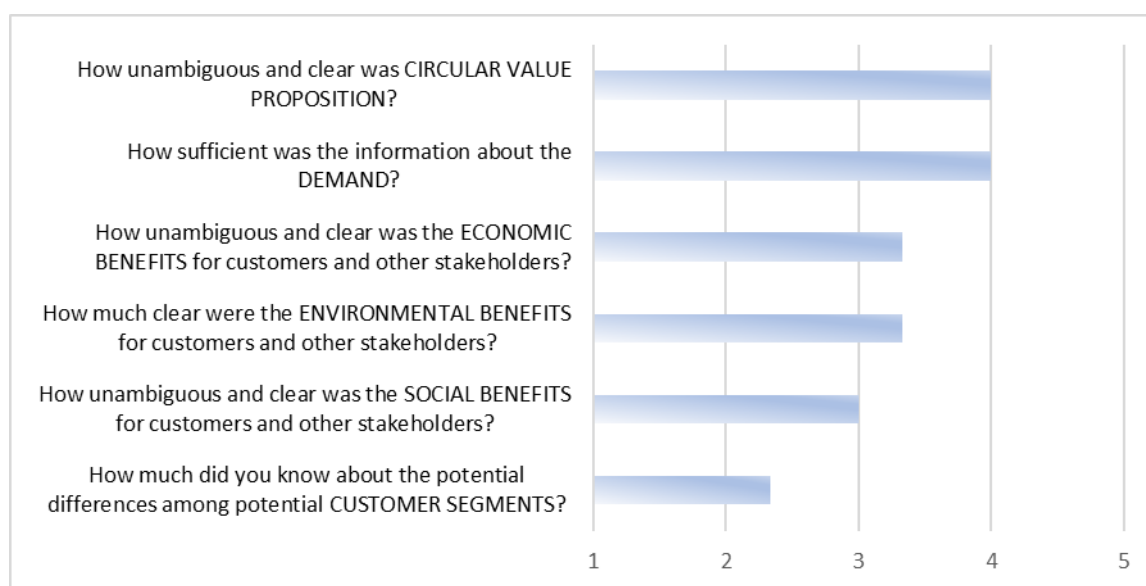


Figure 5: Ambiguity concerning the value and the demand

Attention paid to other Canvas elements

Not all the elements of the canvas' model needed the same intensity of thinking in the initiation phase: on the one hand, not all elements play the same important role in a business model; on the other, some elements might not need extensive modification. From the data obtained, it is apparent that more attention was paid to the key activities - see the following graph capturing the intensity of the weights on each element on a scale of 1 ("we didn't think about it at all") to 4 ("We thought about it, definitely").

The focus on key activities was related to the perceived barriers to the new business model: Two of them were essential: the lack of a suitable washing machine model and a sufficient IT platform - both of these complex systems had yet to be created. Simultaneously, questions were raised about whether such a project could even be completed in the time required.

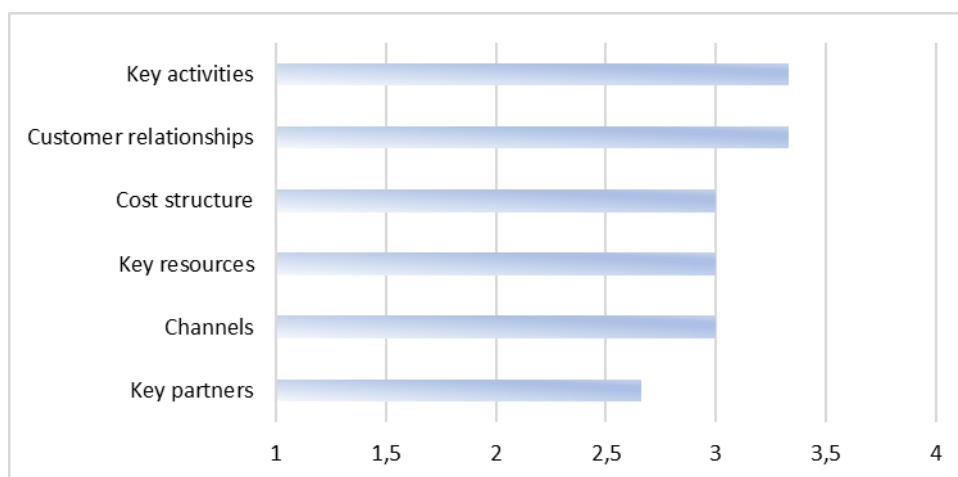


Figure 6: Level of attention paid to selected canvas elements

Organizational aspects

The proposed circular business model represented a significant innovation and challenge for a company that perceived itself (and indeed was) a purely manufacturing company. The novelty and innovativeness of the business model were confronted with a lack of knowledge (including knowledge of the methods required, knowledge of the impact on other products, the associated risk of cannibalization, etc.). From all this, there were also unclear expectations about the whole project. The level of agreement and alignment on the project's objectives was moderate: Whether such a project was or was not aligned to the company's strategic plan. Top management's understanding of the circular model was limited. However, despite various difficulties at this stage, the support from the management was positive. On the other hand, resources given to the project in the initiation phase were limited, as expressed by one of the informants: "we all have to do it on top of everything else".

5.1.2. Ideation phase

Probably due to the relatively clear idea of value offering in the initiation phase, alternative scenarios assessing the value offering (using formalized methods and partially leaning on assumptions) could be developed and evaluated. The choice of the target segment then proved to be a relatively more straightforward task even though sufficient segmentation data was not available initially. As the new circular business model includes new revenues streams, the revenues were assessed/tested and overall profitability right in the early stage of the development.

The main challenge in thinking about the customer relationship was around trust and how to establish new customer relationships and integrate them with other business model elements in general.

The relatively lower attention paid to channels in the first phase was in line with the role of this element in the ideation phase. The view of the channel area reflected the company's desire to make maximum use of the existing distribution system for the new business model - with only a slight modification of its capacity, as no modification of the structure of the distribution system was considered.

Among Key activities design, the area of product development was considered crucial (as there was no such model of washing machine that could be integrated into the circular model). However, significantly more attention was paid to sales and customer relations than before (see also the evaluation in the table below).

Key activities difficult to design because of:	Research & development	Production	Marketing	Sales & customer	IT infrastr.
Complexity					
Missing tools/methods					
Missing data/information					

Note on colours: brown - activity was difficult; light brown - somehow difficult.

Table 10: Perception of difficulties in activities design

Key partners had a similarly unpromised status as channels in the ideation phase.

As far as resources are concerned, in the ideation phase, R&D capabilities and financial resources were the most important, supplemented by marketing capabilities, human resource, partner relations, and logistics capabilities. When comparing how much the resources/capabilities were needed and how much was available, a particular discrepancy emerges for service providing capabilities - they seemed to be necessary, but the company was lacking it somehow.

The whole phase of ideation followed quite smoothly on the first phase of initiation. Although the new business model implies major changes, including perhaps the most difficult of them all - the shift in mindset from a manufacturing company to a service-oriented one - the problems that arose were solved (one could say with average effort – the answers to respective question resulted in the value of 3 on a scale of 1 to 5). The same, somewhat neutral, assessment also applies to the difficulty of the circular model - its complexity was manageable (again, an average rating of 3 on a scale of 1 to 5). In terms of the use of systematic tools, which were mainly used in this phase, it is worth mentioning two of them:

- Process and revenue flows simulation for price setting
- Market research and competition analysis

5.1.3. Integration phase

Moving the project into the integration part did not lead to any difficulties in the interpretation of the value provided. Understanding what value would be provided, to whom (to which stakeholders), when, how and why was the same as in previous phases, i.e., the understanding was clear even though the model itself was elaborated and many details specified – i.e., became more complex.

In terms of the management of the integration phase, some specific tools were used, and several organizational measures occurred, as summarized in the table below. In this context, it is important to mention the following: Firstly, the project considered various risks, even using formalized methods; some mitigation activities were planned for high-impact and high-likelihood risks. Second, although some positions' assigned tasks were changed or expanded, the project did not lead to any structural, organizational change. Thirdly, the attention given to development activities such as staff education and training was only moderate.

Measures and methods		Level of application
mitigation plans for high impact and high-likelihood risks		
risks incorporated into the plans for the implementation phase		
methods for the project risks assessment used		
setting performance indicators for project failure and success		
organizational changes planned for the implementation phase	new department in the organization	
	new position in the organization	
	new tasks for some positions	
	new outsourcing of some activity/process	
	new insourcing of some activity/process	
managerial actions planned for the implementation phase	training and education	
	application of new managerial methods	
	Planned changes in corporate culture	

Note on colours: brown – measure/method applied; light brown – somehow applied; white – not applied.

Table 11: The measures and methods employed in the integration phase

5.1.4. Implementation phase

The following chapter assesses the implementation phase only partially, as this phase is still ongoing and is, therefore, an initial view of the situation.

No new facts have emerged during the implementation phase that would significantly complicate the implementation or require a reconsideration of previous decisions and stages. This does not mean that the project may not have been modified: For practical reasons, the implementation and launch of the demonstration on the Austrian market were abandoned, mainly because the Asko brand is not established there. The Asko brand unfamiliarity on the Austrian market proved to be a relevant barrier. Therefore, this market was replaced by Sweden - the country of origin of the Asko brand. This change naturally means that some of the lessons learned regarding the Austrian market will find the application after the demonstration phase is over, but not now. On the other hand, Gorenje had enough information about the Swedish market; some analyses needed to be completed.

It is worth mentioning other minor issues and topics that required more attention during the implementation. Firstly, more effort was needed for designing the refurbishment activities of used washing machines. Another problem was related to the project's economy, namely, how to secure the funding for the Pay per wash system until the breakeven point would be reached. The letter is linked to the results on the following table (showing that the revenue streams were subject to changes). From the point of view of the demand on the required resources, it has proved necessary to pay more attention to the IT infrastructure (including the IT part of washing machine development, which was very complex and risky). This is connected to delays in updating websites/e-shop in individual markets, which in turns affected the market entry strategy. Lastly, more attention needed was to be paid to taxation and fiscal matters than expected.

An unexpected requirements/information different from the plan	Scale: 1 (extremely much unexpected) - 4 (everything as it was expected)
Circular value proposition	3
Customer segments	3
Customer relationships	3
Channels	3
Key partners	3.3
Key activities	3.3
Key resources	3.3
Cost structure	3.3
Revenue streams	2.6

Table 12: Novelty in canvas elements during the implementation phase

If we compare the integration phase and the actual implementation, we see that there have been no changes concerning the organisation and organisational structure. The data collected only shows that there is probably an increased requirement to change/adjust the corporate culture. Thus, this area will require more detailed attention during the final evaluation of the project, which will take place at a more distant point in time.

Areas of changes		Level of changes
New risk(s) identified		
organizational changes in the implementation phase	new department in the organization	
	new position in the organization	
	new tasks for some positions	
	new outsourcing of some activity/process	
	new insourcing of some activity/process	
managerial actions in the implementation phase	training and education	
	application of new managerial methods	
	Planned changes in corporate culture	

Note on colours: brown – changes in the area; light brown – partial changes; white – no change.

Table 13: Overview of organizational and managerial changes during the implementation

In retrospect, the project was quite different from previous corporate business model innovations as it was not aligned with the current corporate structure and established responsibilities. It was necessary to integrate the circular model into the business operations alongside the existing model and embrace the customers better. Incidentally, gaining deeper customer relationships can be seen as one of the other benefits of this whole innovation. Despite the above incompatibility and the resulting difficulty of the innovation, it was not that difficult to convince essential people in the company of the circular model's meaningfulness. The expected profitability of the project (a new increasing revenue stream after project end) and, at the same time, the support of the European Commission (contribution) also played a significant role in the argumentation. The new model makes the cost structure more flexible.

From employee's perspective, this also had the consequence that the project increased the workload beyond their existing, regular duties and tasks. That was, however, accepted through the help of a positive spirit and leadership.

5.2. Evaluation results for automotive demonstrator

The following assessment is based on the statements of three respondents from C-ECO, supplemented by the views of three Bosch employees representing the remanufacturer's perspective.

5.2.1. Initiation phase

The original idea of the form of circular value in this project was not perfectly clear. Importantly, the lack of clarity concerned the economic value proposition. On the contrary, the degree of certainty related to the environmental benefits that the new circular business model was to deliver was higher. Actually, it was the positive environmental impact that was co-drove the whole innovation project: It was laid on the assumption that the implementation would lead to a significant reduction in the scale of reverse logistics operations and, as a consequence, a reduction in CO2 emissions. Only one respondent could assess the third area of circular value, the social benefits; however, he/she chose a neutral answer on the offered scale of 1 to 5 (circular value was "extremely unclear" to "extremely clear", i.e. a three).

In this context, the unsurprising response was that there was a lack of information about the demand for the value proposition. The notion of demand had only an approximate contour: respondents expressed only a relative sufficiency of information about the demand of some of the customers who would be involved in the new business model. Specifically, the knowledge of particular customer segments was limited. As the innovation aimed at multiple segments, the understanding of segments is crucial.

When evaluating the other business model elements, it appears that the ideas and the intensity of the reflection and thinking about elements set-up were somewhat more precise. On the contrary to the value proposition and knowledge of the segments, the project's experts had more reliable information and data. They did not omit any element in their discussions and paid some or full attention to each remaining business model elements. In numbers: on a four-point scale ranging from 'not at all' to 'completely', the attention was assessed as 3 or 4 (see Fig below).

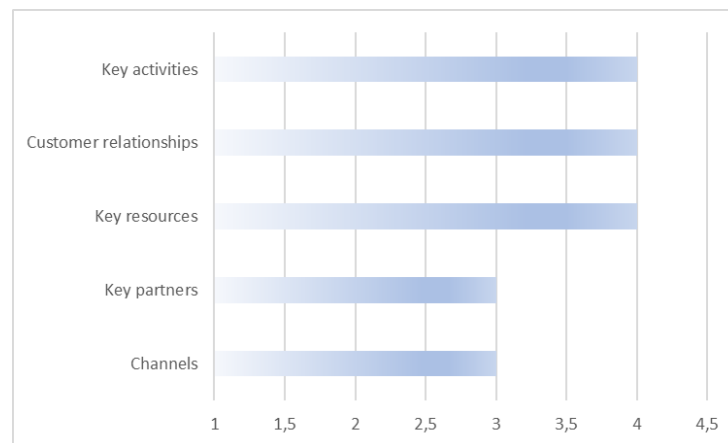


Figure 7: Level of attention paid to selected canvas elements

Next, the responses showed that the lack of resources, whether human, material or financial, was not a central theme at this stage, which is in line with the above findings. The vital question in the initiation phase was whether and how to include other stakeholders in developing the business model and its implementation; that is, the cooperation with external subjects was understood as the success factor of the innovated model.

The lack of management support (which is a common failure reason in many innovations) was not a limitation here: the key managers understood the new model's meaningfulness and alignment with the company's objectives well (as measured by no. 4 on a positive scale of 1 to 5). Moreover, they also showed rather strong support (through leadership and top management commitment) favouring the business model innovation. As a consequence, the new circular business model was embedded in the company's strategy.

There was some discussion about the rationality of the investment into the new model. Whether the in-house cost would pay back over time as the innovation required considerable implicit costs, i.e. the time, energy and know-how that many experts had to devote to the project.

In particular, during the initiation phase, respondents identified the expected complexity of the whole innovation as the barrier. The resulting large number of heterogeneous changes had to be taken into account. The scale of all the changes raised hesitation about whether the innovation (complex and longer-lasting implementation) would ultimately discourage customers from remanufacturing as such.

At the risk of generalization and losing some details, the following can be summarised. The original idea of the new circular business model was clear, mainly in terms of how to set up and implement the new circular business model. It was less clear what effects this solution would have for the different customer segments. In other words, the question of what to offer and whether there would be an adequate demand for such a service from the different market segments (customer groups). Thus, the main challenge the respondents considered was how to communicate the proposed innovation to diverse customers in a clear way and how to explain the benefits from the customer perspective specifically. Communication was seen as the essential factor for the success of the business model innovation.

5.2.2. Ideation phase

The ideation phase did not work much with alternative plans regarding the value provided. Rather, the focus on a single solution resulted from the limited knowledge concerning the exact needs of customers and customer segments. Especially in the first stages of the innovation process, it was not easy to gain customer insights and involve them in the model development process. The co-creation sessions and the interviews with the potential customers turned out to be the most beneficial methods and procedures that gained a better understanding of the customers' position (their gains and pains and their concerns and limits for the new business model).

Thematically, the creative discussions regarding the shape of the customer relationship revolved mainly around the topic of trust: what trust means for customers precisely and what tools to use to support and gain customer trust. The development team worked with the idea that an essential prerequisite for achieving customer trust in the new circular model was the customer understanding of the business domain. This understanding is complicated in that an intangible product is offered; such products are inherently more challenging to describe and measure the quality provided. Further considerations and discussions were given to setting up the model, so that customer privacy was not compromised, another essential and required attribute.

In terms of internal strengths, the following resources and capabilities were played a crucial role in the new business model development, namely "research & development capabilities", "partner relationship", "logistics and supply chain capabilities", "service providing capabilities". Not coincidentally, these were areas that had been developing by the current business model and business activities.

Although there was much to build on (existing business model, capabilities and resource), the new model represented a significant structural change to the company's provided services and business logic. And this particular task - designing systemic change - was quite daunting and required out-of-box thinking. However, the challenge was particularly relevant to the core of the new business model (value delivered, customer relationships and processes). In contrast, the other business model elements were not that difficult to accommodate.

In terms of the methods and tools that were successfully used in the design and change process, the Canvas business model was essentially exclusively used, as many of the necessary sub-steps were quite obvious from the very logic of the business model innovation.

5.2.3. Integration phase

Even in the integration phase, the question of the customer offer was still primarily based on ideas and assumptions, and far from everything was clear. The important point is that the risks were at least partially formulated and included in the implementation plan for the new model at this stage. Similarly, mitigation actions have been identified for some of the risks identified. The question of the success of the innovation was also addressed by defining performance indicators to determine whether the innovation would be a success or failure.

In terms of how the integration phase was managed, it was clear that working on the phase brought new tasks for the staff again. There were no changes in the organizational structure. On the other hand, the integration involved some outsourcing and insourcing; for the employees, the changes included training in new methods and a partial change in the company culture (see table below for an overview).

Measures and methods		Level of application
mitigation plans for high impact and high-likelihood risks		
setting performance indicators for project failure and success		
organizational changes planned in the implementation phase	new department in the organization	
	new position in the organization	
	new tasks for some positions	
	new outsourcing of some activity/process	
	new insourcing of some activity/process	
	new partners	
managerial actions planned for the implementation phase	training and education	
	application of new managerial methods	
	Planned changes in corporate culture	

Note on colours: brown – measure/method applied; light brown – somehow applied; light grey – applied to a very small extent; white – not applied.

Table 14: The measures and methods employed in the integration phase

5.2.4. Implementation phase

During implementation, there has been a slight shift (and additions) in the concept of the circular value proposition. It was necessary to implement the optimization of logistics operations (independent of financial flows), generating additional environmental benefits. It proved to be important to offer not a platform but a service that combines both logistics and IT in terms of economic benefits. The social benefits were affected by the fact that the new model required the integration of more stakeholders than assumed initially.

The extent of necessary modifications to the business models elements that had to be made in the implementation is captured in Tab 15. The assessment was carried out on a scale of 1 to 4, where 1 indicated a major need for modification; lower values together with darker shading indicate elements that required more extensive changes. Given the information uncertainty regarding customer segments (also evident in previous innovation phases), it is not surprising that the customer segments required more extensive adjustments. It is noteworthy that the other areas with more significant adjustments were revenue, while the cost structure relatively needed minimal adjustments.

An unexpected requirements/information different from the plan	Scale: 1 (extremely much unexpected) - 4 (everything as it was expected)
Circular value proposition	3
Customer segments	2.3
Customer relationships	2.6
Channels	2.6
Key partners	3
Key activities	3
Key resources	3
Cost structure	3.3
Revenue streams	2.3

Table 15: Novelty in canvas elements during the implementation phase

There were no new departments or job roles created in the implementation phase either, and so, as in the integration phase, the innovation process was delivered through new tasks for existing job roles. New external partners were heavily involved in the process. In the area of soft management tools, it is worth mentioning a slight adjustment of the corporate culture, namely a stronger focus on dealer and salesman oriented business.

Areas of changes		Level of changes
organizational changes in the implementation phase	new department in the organization	
	new position in the organization	
	new tasks for some positions	
	new outsourcing of some activity/process	
	new insourcing of some activity/process	
	new partners	
managerial actions in the implementation phase	training and education	
	application of new managerial methods	
	changes in corporate culture	

Note on colours: brown – measure/method applied; light brown – somehow applied; light grey – applied to a very small extent; white – not applied.

Table 16: Overview of organizational and managerial changes during the implementation

Within the company's internal environment, the technical interaction between the current and new business model had to be resolved in the context of innovation. However, this was not the main challenge of the implementation. That was the diversity and complexity of the operations themselves, both in terms of IT support for core-accounting processes, logistics and partly technically (defining technical criteria). Overcoming the above challenges was achieved through intensive work and cooperation with project partners. The result is a solution that can be operated and executed on the market. The project team gained profound market knowledge in the implementation

The most significant deviation from the plan that occurred during implementation was the inclusion of wholesalers among direct customers, which was not originally envisaged. Reaching and acquiring wholesalers was, however, highly challenging. What is more, acquiring other necessary stakeholders was similarly problematic. The difficulty was because there were no contacts to these stakeholders, not even through existing networks. Secondly, the new business model had to be explained to them in simple terms, although the model itself is inherently quite complex. Finally, cultural differences were also a complication in attracting stakeholders and expanding of stakeholders' network.

As the result of these adjustments and efforts, vital players in the aftermarket, the core brokers, were gained. For them, the model would facilitate the provision of services to their own customers. Core-brokers themselves should recognize the following positive effects:

- "Reduce their investment risk;
- enable them to serve their customers better;
- trigger relevant core returns to enable more remanufacturing".

In addition to this central challenge, it was also necessary to identify new opportunities that arise from the nature of the business model - the fact that it will be based on options logic. The

development team tried to find solutions (synergies) through analogies with washing machines and other industries.

The risks that were dealt with during the implementation phase related to two areas. The first area reflected the complexity of the project itself (the interconnectedness of individual activities). The second, which is closely linked to the first one, related the IT solutions: specifically, the risks associated with data transfer to/from customers and suppliers, software errors, etc.

5.3. Summary of findings

Following subchapter summarizes major barriers, challenges and risks of circular business model innovation and development process perceived by the experts from demonstrators' companies and from Bosch revealed from survey and from interviews. These are shown in Table 17. The overview of critical areas is presented in following figures.

	CHALLENGES/BARRIERS			RISKS
INITIATION PHASE				
Whole CBM	complexity of the whole innovation	support from people in the company (Gorenje) perception of extra work (Gorenje)	overall lack of knowledge (Gorenje)	feasibility to make product from an idea dependence on other people in the company time business model does not fit to the existing way of doing business
Circular value proposition (CVP)	clarity of the content of CVP (especially economic part – C-ECO, the environmental and social aspects - Gorenje)			preference of ownership before renting
(New) Customer segments	information about the demand for the value proposition	identification of different potential customer segments		
Customer channels				
Customer relationships				
Key partners	whether and how to include other stakeholders in developing the business model and its implementation			

Key activities	how to communicate the proposed innovation to diverse customers in a clear way and how to explain the benefits from the customer perspective specifically	technical development of the washing machines (connected devices) and the entire IT infrastructure		
Key resources	knowledge about new customer segments	top management knowledge of CBM	no IT infrastructure for new CBM	
Cost structure				
Revenue streams	how to generate the revenues			
IDEATION PHASE				
Whole CBM	shift in mindset	out-of-box thinking		
Circular value proposition (CVP)				
(New) Customer segments				
Customer channels				
Customer relationships	how to establish new customer relationships			
Key partners				
Key activities	sales and customer relationship management design service providing capacities and capabilities	involvement of customers in the model development process	building the trust of customers	
Key resources	customer insights			
Cost structure				
Revenue streams				
INTEGRATION PHASE				
Whole CBM			missing “big picture” of the project and its impacts and of the business case	knowledge of the future market conditions new business
Circular value proposition (CVP)	detailed content of CVP			
(New) Customer segments				
Customer channels				
Customer relationships				contracts

Key partners				power relationships
Key activities				
Key resources				
Cost structure				
Revenue streams				
IMPLEMENTATION PHASE				
Whole CBM	diversity and complexity of the operations and complexity resulting from variety of core-accounting process, different technical criteria, logistics processes over various remanufacturers	time	alignment of the new CBM with the current corporate structure and established responsibilities to manage technical interaction between old logic and new approach	mastering complexity of CBMI time to transfer problems into learning and improvement lack of data transmission from customers' supplier or from customer to allow for smooth supply chain de-prioritized implementation for customer software issues
Circular value proposition (CVP)	adaptation of intended value proposition (product is not IT platform but a bundle of services that bring expected CVP)	enlargement of CVP (more social benefits because of the involvement of more stakeholders in comparison to plan)	understanding of customers' perception and expectations of CVP	functioning product
(New) Customer segments	acquisition (no existing contacts) unfamiliarity with brand	cultural barriers	communication (gap between complexity of CBM and project and expectation of simple explanation)	entering new market
Customer channels				
Customer relationships				

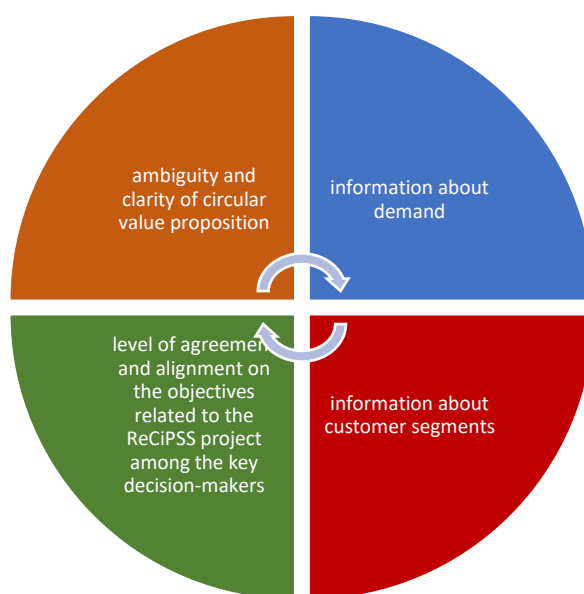
Key partners	the same as with new customer segments			finding new and right partners
Key activities	accounting designing the refurbishment activities of used washing machines IoT development	pricing		
Key resources	necessity to create database of standardised core information (which unified the different ways of records and evidence of different actors) IT infrastructure	to secure the funding for the Pay per wash system until the breakeven point is reached	knowledge of taxation and fiscal matters in different markets	
Cost structure				
Revenue streams	adaptation of revenues streams			

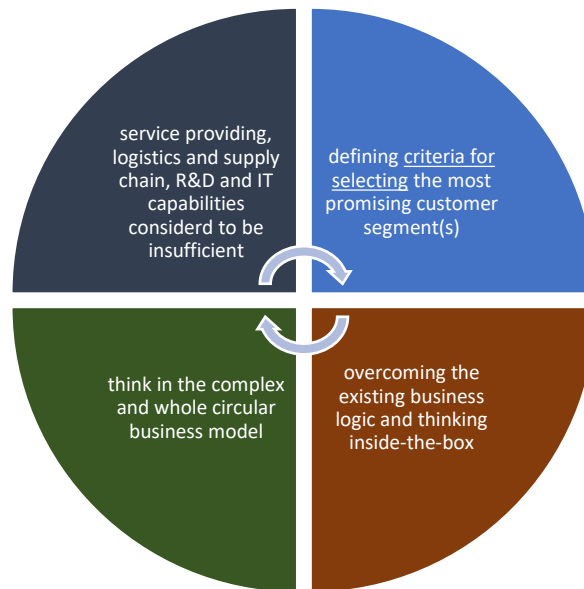
Table 17: Summary of barriers, challenges and risks

Key critical areas

In every phase of the circular business model innovation process some critical areas were detected. All of them could serve as risks for the companies who decide to innovate their business model for circular economy.

Initiation phase



Ideation phase*Integration phase*

In this phase only one critical area has been discovered, specifically the knowledge of what circular value proposition for customers and other stakeholders will be created, delivered and captured by whom how, where, when and why. In this case only vague assumptions and ideas existed.

Implementation phase

6. Guidelines for the systematic risk assessment

Systematic risk assessment requires an iterative Plan-Do-Check-Act approach. Three basic processes create this first three parts of this approach, as shown in the following figure (based on Altuğ, 2002). Assessment should be relatively simple for not overload managers and employees with extrawork, however, relevant to the complexity of the object of assessment (Marle and Gidel, 2014).



Figure 8: Three basic processes of the iterative approach for systematic risk assessment

1st process: Risk management planning

Systematic risk assessment should start before any decision to start with circular business model innovation is made. It means during the initiation phase relevant persons must think about potential events which might influence success of the intended innovation and about existing uncertainties and to plan concrete further steps concerning risk management. The more radical the innovation is the more the need for formal plan exists.

2nd process: Risk identification

For circular business model innovation, we recommend identifying possible risks in four phases (1) Initiation; (2) Ideation; (3) Integration; (4) Implementation, and relevant risk areas or sources (see chapter 3.4 and 3.5 for some more information about the character of risks and process of risk identification).

The result of the risk identification is a **Risk matrix** (see Table 18 below as an example or as a pool of risks) combining 4I Framework and several categories or areas/sources of risks. The defined risks are based on desk research of various authors (see, e.g., Bianchini et al., 2019; Bocken and Geradts, 2020; Govindan and Hasanagic, 2018; Linder and Williander, 2017; Vermunt et al., 2019, etc.), literature review presented in this Deliverable and results of the interviewing. The risk matrix is adaptable – it could be filled in with any other relevant risk, and the other areas relevant to the risk assessment may be added. As shown in chapter 3.4, risks can be numbered for better risks evidence.

	Initiation	Ideation	Integration	Implementation
Economic and financial risks	Identification of relevant key resources (material, finance, people)	A potential problem with setting the specific expense and costs of CBM	Lack of operational resources	Long "pay-back period"
	Resources capacity problems - financial, sourcing	Higher uncertainties regarding financial value creation	Return flow challenge	Capital tied-up (especially in PSS)
	Investment risk	Need for high long-term investments	Financial risk transfers from the customer to the producer	Additional costs of operating a CBM
			Costly management and planning processes due to more complex practices	Right prices bringing revenues
Social, political, and environmental risks	Fear of failure - the potential of a CBM idea	Identification of relevant new types of relationships, new channels, new key partners, new key activities	Fear of integration failure	Fear of implementation failure
	Fear of cannibalization of the existing business model	Fear of failure - during the process	Resources capacity risks - partners relationship	Failure in customer relations regarding trust issues (speed of reaction, honesty, security, privacy, friendliness, ...)
	A cultural shift from an engineering- and product-centred core culture to a more relational and customer-oriented one	Unexpected changes in key partners' requirements/need when creating circular value	Unexpected changes in key partners' requirements/need when capturing circular value	Demand volatility
	Geographical dispersion	Insufficient knowledge of target customer segments	An adverse change to legislation that has a significant impact on CBM	Non- or low acceptance of value proposition from customers
	Rigidity in customer behaviours and business routines	Non-compliance with applicable laws and regulations	Unexpected discovery about customer segments	Pressure on more intensive information and data sharing
	The complexity of regulations, lacking		Unexpected requirements on additional service	Higher dependence on external parties

	conducive legal system and poor institutional framework			
			Unexpected changes in key partners' requirements/need when delivering circular value	Lack of supply network support
Technical and process-related risks	Ambiguous and unclear value proposition (economic, social, environmental benefits)	Insufficient information about the demand	Demands on ICT based platform	Unexpected changes in process structure when delivering circular value
	No existing methods and/or methodology on how to proceed	Insufficient data and information	Compatibility of organizational changes	Lack of technological expertise (repairability, remanufacturing)
	Compatibility of new CBM with a strategic plan	Resources capacity problems - IT & ICT, logistics and supply chain, production capabilities, service providing capabilities	Lack of channel control	Lack of IT infrastructure
	Conflict of interests between the two business models	Changes in value generation	Need for technical and technological know-how and expertise	Adoption of specific technologies for the redesign of circular products and production systems
Personnel, organizational and managerial risks	Lack of enthusiasm and commitment of key persons, leadership, and top management	Lack of enthusiasm/commitment of key persons, leadership, and top management	Lack of enthusiasm/commitment of key persons, leadership, and top management	Lack of enthusiasm/commitment of key persons, leadership, and top management
	Lack of experience and/or knowledge	Lack of understanding of the idea from top managers	Resources capacity problems - R&D capabilities, human resources, marketing capabilities	Difficulties with project failures and success measuring
	Resistance during the whole process	Risk of overcoming the existing business logic	Difficulties with project failures and success measuring	Resistance during the whole process
	Need for new organizational competencies (e.g., team motivation)	Difficulties with thinking out-of-the-box for the CBM	Resistance during the whole process	Organizational capabilities necessary for implementing CBM across different organizational functions

		Difficulties in the complex CBM thinking	Cross-department collaboration	Maintaining the same quality level
		Resistance during the whole process	Corporate bureaucracy	Conflicting interests
		Short-termism	Staff shortages	Competence and power structure destruction
		Shortage of time for the evaluation and selection of ideas	Conflicting interests	

Table 18: Circular business innovation risk matrix

Another Risk matrix can involve different dimensions, as shown in Table 19 that reflects risks categorized according two dimensions – elements of circular business model canvas and phases of circular business model innovation or development.

CBM canvas elements	Phases of CBMI			
	Initiation	Ideation	Integration	Implementation
Circular value proposition	Risk... Risk...			
Customer segments				
....		Risk...		
....			Risk...	
Revenue streams and circular economy benefits				

Table 19: Risk matrix – phases of CBMI and CBM canvas elements

3rd Process: Risk analyses

Except simple risk assessment (see chapter 3.4) in case of circular business model risk analyses we recommend working with the factor of impact (or consequence) and with the factor of probability. The simplest way is to use suitable ranking of the factors, so the scale reflects the character of both factors (for the example of ranking see chapter 3.4).

Interdependencies between the risks emerging as the consequences of risk occurrence in one element of CBM in the other CBM elements should be analyzed and evaluated as well. It could be done through brainstorming and example of result showing such interdependencies is shown in Table 20 and in form of a map using business model canvas elements in Table 21.

Source(s) of risk(s)	Identified risk(s)	Impact of risk on.....(part of CBM element)	Description of impact	Description of secondary impact	Description of tertiary impact
suppliers	high bargaining power of suppliers of spare parts	Circular customer value	Price increase of spare parts	Needed change of contract conditions	Dissatisfaction of customer

Table 20: Interlinkages between source of risk, risk and impact of risk

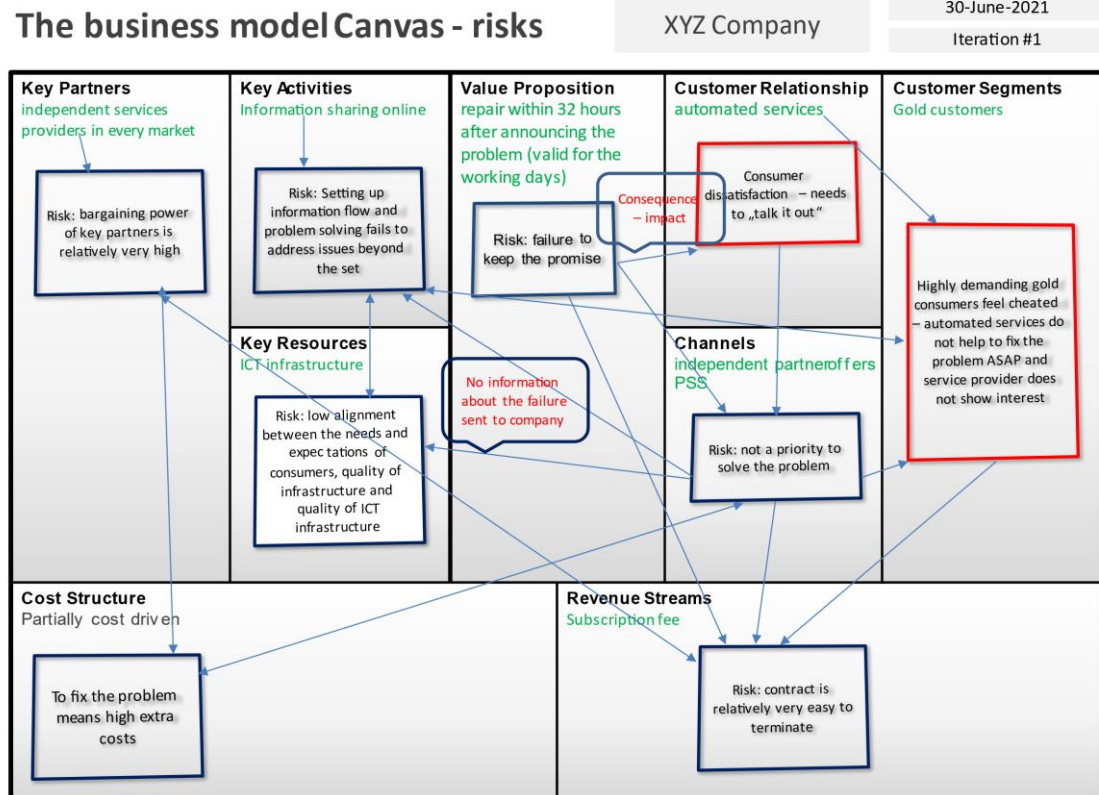
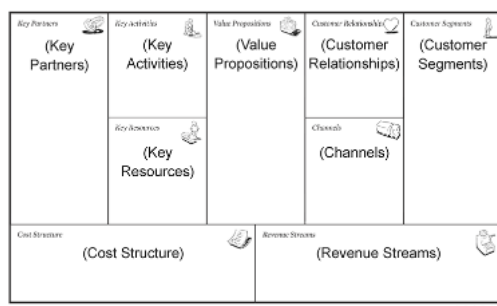


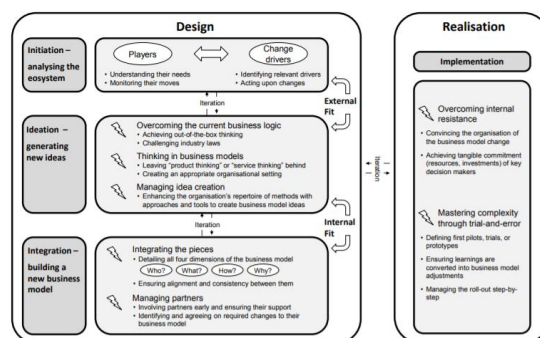
Table 21: Interdependencies between the risks (and consequences) within business model canvas elements

For the analysis we recommend also more structured process as well as risk maturity assessment tool.

Following figure represent logic of suggested process of the risk analysis



Source: www.businessmodelgeneration.com



Source: Frankenberger et al, 2013

	Initiation	Ideation	Integration	Implement.
Value Propositions				
Customer Segments				
Channels				
Customer Relationships				
Revenue Streams				
Key Activities				
Key Resources				
Key Partnerships				

Each canvas element is analyzed in the initiation, ideation, integration, and implementation phase separately in a semiquantitative and qualitative way.

Initiation						
		Clear circular value proposition (economic)	Clear circular value proposition (environ.)	Clear circular value proposition (social)	Sufficient information about the demand	Value proposition (3.00 out of 5)
Circular value proposition	++	0,1	1	0	0	0
	+	0,2	0	0	0,2	0,2
	o	0,1	0	0,5	0,6	0,6
	-	0,5	0	0,5	0,2	0,2
	--	0,1	0	0	0	0
Identified challenges and problems						
Lack of supporting regulations						
Fear of the unknown						
Lack of methods for handling life cycle ...						

Figure 9: Logic of risk analysis process

The answers received from experts are graphically presented as relative frequencies (in bar charts), showing the distributions of answers on the given scale one (depicted as --) to five (++) - see the example above concerning the Initiation and circular value proposition. The overall evaluation of each canvas element in a given innovation phase is computed using the arithmetic average (the number 3 in the example above). The evaluation of canvas elements in such a quantitative way enables a benchmarking of business model elements across four innovation phases (as depicted in Figure 4), visualizing the potential gaps in the end.



Figure 10: Sample Picture/Drawing

The same logic is applied also for the maturity and risks assessment.

7. Conclusions

This Deliverable is the interim evaluation report of both demonstrators' circular business model development and innovation involved in the ReCiPSS project.

The main impact of this Deliverable is to bring an overview of barriers, challenges, and risks of managing business model innovation for the circular economy and circular Product-Service systems. Such an overview may help demonstrators be more aware of what to avoid and pay more attention to and care of in the following phases of implementing a circular business model. It also can be helpful for the external readers from companies and navigate them through the business model innovation process on their route to the circular economy. In addition to an overview, it also guides systematic risks assessment associated with the circular business model development and innovation.

Findings from the evaluation both confirm existing knowledge of issues of business model innovation which are the same for circular business model innovation, as for instance risk of organizational resistance, risk of the consequences of insufficient knowledge, organizational culture that inhibits innovation, insufficient resources etc. It also confirms risk of the alignment between existing linear business models and new circular model. Evaluation show differences and similarities in the innovation process between the company who is on the path of radical innovation (develops new circular business model from scratch) and company which only change some features of several elements of already existing model with some aspects of circularity.

Review and evaluation also expand current knowledge of circular business model innovation. It summarizes barriers, challenges, and risks of this kind of innovation and broaden the understanding. Findings stressed the need to recognize the role and peculiarities of service provision in product services systems and specific features of IT services. Misunderstanding or neglecting of this issue could be very risky for success of the circular business model innovation that is highly dependent on IT and IS. It also reveals how accounting, design and standardization need to be adapted for circular economy and highlights the complexity of CBM and CBMI. Furthermore, it also shows that project approach is probably restrictive for CBMI because time and deadlines of milestones that are typical for project management also represent a barrier. As literature argues, CBMI is process of learning and experimenting where the road to success goes through failures and dead-end road or through trial and error.

Reports in this Deliverable have several limitations, some of which may also act in the self-evaluation or in the further evaluation by another party. First, covid and some project evolution related barriers and problems with covid pandemic situation negatively influenced time when important processes have started (implementation of models) with the consequences on the evaluation. Pandemic also resulted in the distant evaluation, although the personal presence and on-site visits and face-to-face interviews would be probably more time-efficient and more effective. Second, the non-existence of an example of the evaluation methodology led to developing a document, which should be tested and improved. In particular, the questionnaire should be revised and simplified, which is rather difficult task if the complexity of circular business model innovation should be the covered. Formulation of right questions is a challenge. Third, evaluation of circular business model innovation to be robust needs cooperation and willingness to participate in the evaluation from many in-project involved persons across the company and partners' organization. This is an extremely demanding organizational task and requirement that needs additional time and effort from company-level project members to

explain the evaluation's relevance and push and convince people to participate. Fourth, for the robust evaluation, more or multiple sources of evidence (for instance, not only interviewing but also observation and a kind of the ethnographic and/or participatory research, content analysis of documents, e.g. existing analyses, performance measurement reports) would be very beneficial. Fifth, evaluation – as the iterative and continuously improved process – should be repeated.

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Appendixes

Appendix A: Bosch questions

What were the main drivers to participate in the project?

What was the purpose – objectives - of your participation in the project?

What do you perceive as the most risky with the project?

What were and are major challenges of the project?

Were there any analyses done before the project?

Are there any changes associated with the project?

Are there any new resources needed for the project?

Were there any measures prepared and applied for the project?

Do you evaluate now how the project is going on?

Are there any changes in the external environment which positively or negatively affect the implementation of the ICT portal?

Appendix B: CECO questionnaire

Questionnaire:

MAPPING THE PROCESS OF THE CIRCULAR BUSINESS MODEL DEVELOPMENT

The questionnaire maps **the managerial and organizational aspects** of the development process of the circular value *creation, delivery and capture*.

The answers will enable an evaluation of the whole process of the circular business model development (introduction of the ICT platform into the reverse logistics services) and help to identify potential improvements and countermeasures which become part of guidelines for systematic risk assessment and transition management in the implementation process of circular product systems.³

Information to the questionnaire and instructions

Questions cover the following thematic areas:

1. a basic respondent identification;
2. the stages of the innovation process;
3. in some cases, elements of the business model canvas⁴;
4. changes during the development process (assumptions vs reality) - the risks and the impact of change.

Most of the questions are in the form of yes-no or scale answers; some are open – we will be delighted if you add some words there.

Please **SKIP questions that are not RELEVANT FOR YOU** (e.g. if YOU DID NOT PARTICIPATE in the initiation, ideation, integration or implementation phase OR YOU ARE NOT FAMILIAR with some elements of the business model canvas.)

To fill in the questionnaire, you need to download it and open in MS Word (not in web/office live version). Most of the questions are in the form of single (yes/no) and multiple choice; please tick the appropriate box(es) ☐. Scale answers are to choose from the drop-down menu (see the picture below). Several questions are open-ended – we will be delighted if you add some words by clicking and editing the text field (see the example below) – you start to write the text simply to the field.

³ Questionnaire is a part of the Task 2.4 and answers are the input into the Deliverable 2.3

⁴ Business model canvas is a tool of strategic management to design, redesign and visualize business models. It contains 9 elements – building blocks that concern all necessary parts of the processes of value creation, value delivery and value capture. More information here: https://en.wikipedia.org/wiki/Business_Model_Canvas

1. How **unambiguous and clear** was the **circular value proposition** (e.g. the value offered to customers through the new circular business model) when the initial idea for pay per wash emerged?

Choose an appropriate item:

Choose an appropriate item.

1 extremely ambiguous and unclear

2 rather ambiguous and unclear

3 neither/nor

4 rather unambiguous and clear

5 extremely unambiguous and clear

How much clear were the economic benefits for customers and other stakeholders?

How much clear were the environmental benefits for customers and other stakeholders?

How much clear were the social benefits for customers and other stakeholders?

3. What was the biggest challenge in the initiation phase concerning value proposition?

Click or tap here to enter text

Your involvement and commitment are needed due to finding out opinions and experiences from your company's internal environment.

Thank you very much for your collaboration. We highly appreciate your willingness to answer the questions.

Masaryk university team

Masaryk university team

Information about the respondent

Name and surname:

Position in the company:

How long have you been in this position? Since (the year, month):

Initiation – relevant for persons who participated in this phase

Circular business model canvas:

1. How **unambiguous and clear** was the **circular value proposition** (e.g. the value offered to customers through the adapted circular business model) when the initial idea for the ICT platform and circular logistics services emerged?

Choose an appropriate item:

How much clear were the economic benefits for customers and other stakeholders?

Choose an appropriate item:

How much clear were the environmental benefits for customers and other stakeholders?

Choose an appropriate item:

How much clear were the social benefits for customers and other stakeholders?

Choose an appropriate item:

2. How **sufficient** was the **information about the demand** for such value proposition in this initiation phase?

Choose an appropriate item:

3. Did you think about new types of relationships for circular value creation, delivery and capture in the ideation phase?

Choose an appropriate item:

4. Did you think about new channels for circular value creation, delivery and capture⁵ in the ideation phase?

Choose an appropriate item:

5. Did you think about new key partners for circular value creation, delivery and capture in the ideation phase?

Choose an appropriate item:

6. Did you think about new key activities for circular value creation, delivery and capture in the ideation phase?

Choose an appropriate item:

7. Did you think about new key resources for circular value creation, delivery and capture in the ideation phase?

Choose an appropriate item:

8. In which way was the current or expected existence/lack/insufficiency of key resources (such as material, finance, people) inspiring for the project – and the circular business model?

Click or tap here to enter text.

Additional questions:

1. What was the level of agreement and alignment on the objectives related to the ReCiPSS project among the key decision-makers?

Choose an appropriate item:

⁵ In circular economy, value is often co-created by several partners, delivered and captured by these partners, who may be in the position of customers and suppliers at the same time

2. *If the agreement/alignment was extremely or rather low, what were the reasons? (tick all relevant answers):*

☐ Fear of failure

☐ Insufficient data and information. Please specify what kind of data and information the biggest issue was?

Click or tap here to enter text.

☐ Lack of experience. Please identify what kind of experience was lacking?

Click or tap here to enter text.

☐ Lack of knowledge – please identify what kind of knowledge was lacking?

Click or tap here to enter text.

☐ Money

☐ Lack of understanding of the idea from top managers

☐ No existing methods and/or methodology on how to proceed

☐ Fear of cannibalization (existing business model)

☐ other. Please, specify

Click or tap here to enter text.

3. *What was/were the biggest difference(s) of opinions in key persons?*

Click or tap here to enter text.

4. *What role did leadership and top management commitment play in the process of the circular business model (based on the ICT platform) initiation?*

Choose an appropriate item:

5. *How was the new circular business model involved in the existing strategic plan of the company?*

Choose an appropriate item:

6. *What environmental drivers have inspired you for the circular business model development?*

Click or tap here to enter text.

7. *What was the biggest perceived risk(s) in the initiation stage?*

Click or tap here to enter text.

8. *Which biggest barrier(s) was perceived in the initiation stage?*

Click or tap here to enter text.

9. *What was the biggest challenge in the ideation phase concerning the customer circular value proposition?*

Click or tap here to enter text

Ideation – relevant for persons who participated in this phase

Circular business model canvas:

1. *If you developed scenarios for the ICT platform-based model, how did you assess the scenarios (the alternative plans)?*

Choose an appropriate item:

2. *Which aspect of customer relations did you find most crucial for the business model success in the ideation phase?*

☐ trust ☐ speed of reaction ☐ honesty ☐ security ☐ privacy ☐
friendliness ☐ other, please, specify:.....

3. *Which methods and tools turned out to be the most valuable and successful in key activities (re)design and development? Which activities were difficult to (re)design because of missing or inappropriate methods/tools? For which activities (re)design was the lack of data most crucial?*

Click or tap here to enter text

4. *Which of your resources/ capabilities did you perceive as essential for the decision to undergo the circular business model based on the ICT platform?*

☐ research&development capabilities ☐ human resources ☐ partners relationship
☐ financial IT and ICT ☐ logistics and supply chain resources (tangible) ☐ logistics and
supply chain capabilities ☐ marketing capabilities ☐ production capabilities ☐ service
providing capabilities ☐ sourcing capabilities ☐ other...if yes, please, specify:.....

Additional questions:

1. *How many problems there were with overcoming the existing business logic and with thinking out-of-the-box for the circular business model based on the ICT platform?*

Choose an appropriate item:

2. *How difficult was it to think in the complex and whole circular business model (in comparison to the business model you are used to work with)?*

Choose an appropriate item:

3. *If yes, what were the most severe difficulties?*

Click or tap here to enter text.

4. Which more or less systematic tools to develop ideas concerning *circular business model based on the ICT platform* were applied in the ideation stage if any? If not, why not?

Click or tap here to enter text.

Integration – relevant for persons who participated at this phase

There is no specific question for this phase concerning individual business model canvas building blocks

How much it was clear and became a part of plans what circular value proposition for customers and other stakeholders will be created, delivered and captured by whom (also consider the key partners), how, where, when and why?

Choose an appropriate item:

1. *Have you formulated mitigation plans for high impact and high-likelihood risks?*

Choose an appropriate item:

2. *Have you prepared any performance indicators for the evidence of the project failure and success?*

Choose an appropriate item:

3. *Have there been any necessary organizational changes incorporated into the plans for the business model implementation?*

New position in the organizational structure?

☐ Yes ☐ No

New department in the organizational structure?

☐ Yes ☐ No

New tasks for some positions?



☐ Yes ☐ No

New outsourcing of some activity/process?

☐ Yes ☐ No

New partners?

☐ Yes ☐ No

New insourcing of some activity/process?

☐ Yes ☐ No

4. *Have there been any necessary managerial issues incorporated into the plans for the circular business model implementation?*

Training and education

☐ Yes ☐ No

New methods of management (for planning, organizing, controlling, motivating, communication etc.)

☐ Yes ☐ No

Need for some changes of corporate culture?

☐ Yes ☐ No

5. *Have there been any risks incorporated into the plans for the circular business model implementation?*

☐ Yes ☐ No

6. *If yes, have there been any methods for the project risks assessment involved?*

☐ Yes ☐ No

Implementation – relevant for persons who participated at this phase

Circular business model canvas:

1. *What elements of the circular value proposition have to be added/adapted/changed during the implementation stage? Think about the environmental, economic and social elements.*

Environmental:

Click or tap here to enter text.

Economic

Click or tap here to enter text.

Social

Click or tap here to enter text.

2. *Is there any new, unexpected information about your customer segments you are discovering during the implementation stage?*

Choose an appropriate item:

3. *Are there any unexpected requirements/needs to build and maintain new ways for customer relationships (e.g. additional service – care, additional information etc.) - different from the plan?*

Choose an appropriate item:

4. *Are there any unexpected changes in channel structure, channel members' requirements/needs when delivering circular value to customers – different from the plan?*

Choose an appropriate item:

5. *Are there any unexpected changes in key partners' requirements/needs when creating, delivering and capturing circular value – different from the plan?*

Choose an appropriate item:

6. *Are there any new partners for circular value creation, value delivery and value capture – different from the plan?*

Choose an appropriate item:

7. *Are there any new, unexpected key activities for circular value creation, value delivery and value capture – different from the plan?*

Choose an appropriate item:

8. *Are there any new, unexpected, needed key resources for circular value creation, value delivery and value capture – different from the plan?*

Choose an appropriate item:

9. *Have been some changes in the cost structure needed during the implementation of the circular business model – not expected in the plan?*

Choose an appropriate item:

10. Have some new revenue streams or some changes in the revenue streams occurred during the implementation stage? (speed, amount, character etc.)

Choose an appropriate item:

Here you can write your comments, ideas to the individual building blocks:

Click or tap here to enter text.

Additional questions:

1. Have there been any necessary organizational changes incorporated into the circular business model implementation – different from the plan?

New position in the organizational structure?

☐ Yes ☐ No

New department in the organizational structure?

☐ Yes ☐ No

New tasks for some positions (e.g. marketing, production, logistics, sourcing etc.)?

☐ Yes ☐ No

New outsourcing of some activity/process?

☐ Yes ☐ No

New partners?

☐ Yes ☐ No

New insourcing of some activity/process?

☐ Yes ☐ No

Otherwhat specifically?.....

2. Have there been any necessary managerial issues incorporated into the circular business model implementation - different from the plan?

Training and education

☐ Yes ☐ No

New management methods (for planning, organizing, control and controlling, motivating, communication etc.)

☐ Yes ☐ No

Need for some changes of corporate culture?

☐ Yes ☐ No

Otherwhat specifically?.....

☐ Yes ☐ No

3. *Are there any new risks with the implementation of the circular business model – different from the plan?*

☐ Yes ☐ No

Changes

What are:

1. *the biggest changes during the implementation phase compared to the initial assumptions and plans (do not mention covid-19)?*

Click or tap here to enter text.

2. *the biggest challenges/barriers during the implementation phase compared to the initial assumptions and plans (do not mention covid-19)? How were these challenges solved, and with what effects?*

Click or tap here to enter text.

In both cases, please, describe them individually, if possible:

- a) Circular value proposition
- b) Customer segment
- c) Customer relationships
- d) Channel
- e) Key partners
- f) Key activities
- g) Key resources
- h) Cost structure
- i) Revenue streams

3. *Do you consider the circular business model development process different from other business model development or innovation processes in the company?*

Choose an appropriate item:

If yes, what was different and why?

Click or tap here to enter text.

4. *Are there any challenges related to the simultaneous management of the existing business model and the new model (ICT platform and logistics services)? Any conflict of interests between the two business models, for example?*

☐ Yes ☐ No

If yes, what challenges?

Click or tap here to enter text.

5. *How did you overcome internal but also external (key partners) resistance during the whole process – from the initiation till the implementation?*

Choose an appropriate item:

If there was some internal or external resistance, what was the problem?

Click or tap here to enter text.

6. *Are there any new beneficiaries – new recipients – stakeholders of the circular value your circular business model generates not planned from the initiation phase?*

☐ Yes ☐ No

If yes, who and what value they obtain?

Click or tap here to enter text.

Risks and impact of change

1. *What are the biggest risks during the implementation phase compared to the initial assumptions and plans (do not mention covid-19)?*

Click or tap here to enter text.

2. *Has an impact value chain (input – activity – output – outcome – impact) been created?*

☐ Yes ☐ No

If not, why not?

Click or tap here to enter text.

3. *How do you assess the impacts of the change process? Ex-ante or ex-post or during?*

☐ Ex-ante ☐ ex-post ☐ during

4. *Do you use any proxy indicators to assess the impact in monetary terms?*

☐ Yes ☐ No

If so, which ones and for which activities?

Click or tap here to enter text.

Appendix C: Gorenje questionnaire

Questionnaire:

MAPPING THE PROCESS OF THE CIRCULAR BUSINESS MODEL DEVELOPMENT

The questionnaire maps **the managerial and organizational aspects** of the development process of the circular value *creation, delivery and capture*.

The answers will enable an evaluation of the whole process of the circular business model development and help identify potential improvements and countermeasures that become part of guidelines for systematic risk assessment and transition management in the implementation process of circular product systems.⁶

Information to the questionnaire and instructions

Questions cover the following thematic areas:

5. a basic respondent identification;
6. the stages of the innovation process;
7. elements of the business model canvas⁷.
8. Changes during the development process (assumptions vs reality) - the risks and the impact of change.

Most of the questions are in the form of yes-no or scale answers; some are open – we will be delighted if you add some words there.

Please **SKIP questions that are not RELEVANT FOR YOU** (e.g. if YOU DID NOT PARTICIPATE in the initiation, ideation, integration or implementation phase OR YOU ARE NOT FAMILIAR with some elements of the business model canvas.)

To fill in the questionnaire, you need to download it and open in MS Word (not in web/office live version). Most of the questions are in the form of single (yes/no) and multiple choice; please tick the appropriate box(es) ☐. Scale answers are to choose from the drop-down menu (see the picture below). Several questions are open-ended – we will be delighted if you add some words by clicking and editing the text field (see the example below) – you start to write the text simply to the field.

⁶ Questionnaire is a part of the Task 2.4 and answers are the input into the Deliverable 2.3

⁷ Business model canvas is a tool of strategic management to design, redesign and visualize business models. It contains 9 elements – building blocks that concern all necessary parts of the processes of value creation, value delivery and value capture. More information here: https://en.wikipedia.org/wiki/Business_Model_Canvas

1. How **unambiguous and clear** was the **circular value proposition** (e.g. the value offered to customers through the new circular business model) when the initial idea for pay per wash emerged?

Choose an appropriate item:

Choose an appropriate item.

1 extremely ambiguous and unclear

2 rather ambiguous and unclear

3 neither/nor

4 rather unambiguous and clear

5 extremely unambiguous and clear

How much clear were the economic benefits for customers and other stakeholders?

How much clear were the environmental benefits for customers and other stakeholders?

How much clear were the social benefits for customers and other stakeholders?

3. What was the biggest challenge in the initiation phase concerning the value proposition?

Click or tap here to enter text

Your involvement and commitment are needed due to the necessity to find out opinions and experiences from your company's internal environment.

Thank you very much for your collaboration. We highly appreciate your willingness to answer the questions.

Masaryk university team

Information about the respondent

Name and surname:

Position in the company:

How long have you been in this position? Since (the year, month):

Initiation – relevant for persons who participated in this phase

Circular business model canvas building blocks:

1. How **unambiguous and clear** was the **circular value proposition** (e.g. the value offered to customers through the new circular business model) when the initial idea for pay per wash emerged?

Choose an appropriate item:

How much clear were the economic benefits for customers and other stakeholders?

Choose an appropriate item:

How much clear were the environmental benefits for customers and other stakeholders?

Choose an appropriate item:

How much clear were the social benefits for customers and other stakeholders?

Choose an appropriate item:

2. How **sufficient was the information about the demand** for such value proposition in this initiation phase?

Choose an appropriate item:

3. What was the biggest challenge in the initiation phase concerning the customer circular value proposition?

Click or tap here to enter text

4. How much did you know about the potential differences among potential customer segments relevant for the circular business model in the initiation phase?

Choose an appropriate item:

Please, add your comment if necessary:

Click or tap here to enter text

5. Did you think about new types of relationships for circular value creation, delivery and capture in the initiation phase?

Choose an appropriate item:

6. Did you think about new channels for circular value creation, delivery and capture⁸ in the initiation phase?

Choose an appropriate item:

7. Did you think about new key partners for circular value creation, delivery and capture in the initiation phase?

Choose an appropriate item:

8. Did you think about new key activities for circular value creation, delivery and capture in the initiation phase?

Choose an appropriate item:

9. Did you think about new key resources for circular value creation, delivery and capture in the initiation phase?

⁸ In circular economy, value is often co-created by several partners, delivered and captured by these partners, who may be in the position of customers and suppliers at the same time

Choose an appropriate item:

10. In which way was the current or expected existence/lack/insufficiency of key resources (such as material, finance, people) inspiring for the project – and the circular business model?

Click or tap here to enter text.

11. Did you think about the potential problems with setting the specific expenses and (fixed/variable) cost of the new circular business model in the initiation phase?

Choose an appropriate item:

12. Was the cost reduction one of the circular business model drivers in the initiation phase?

Choose an appropriate item:

13. Was the potential revenue stream from the circular business model a motivator in the initiation phase?

Choose an appropriate item:

Here you can write your comments, ideas to the individual building blocks:

Click or tap here to enter text.

Additional questions:

10. What was the level of agreement and alignment on the objectives related to the ReCiPSS project among the key decision-makers?

Choose an appropriate item:

11. If the agreement/alignment was extremely or rather low, what were the reasons? (tick all relevant answers):

☐ Fear of failure

☐ Insufficient data and information. Please specify what kind of data and information the biggest issue was?

Click or tap here to enter text.

☐ Lack of experience. Please identify what kind of experience was lacking?

Click or tap here to enter text.

☐ Lack of knowledge – please identify what kind of knowledge was lacking?

Click or tap here to enter text.

- ☐ Money
- ☐ Lack of understanding of the idea from top managers
- ☐ No existing methods and/or methodology on how to proceed
- ☐ Fear of cannibalization (existing business model)
- ☐ other. Please, specify

Click or tap here to enter text.

12. *What was/were the biggest difference(s) of opinions in key persons?*

Click or tap here to enter text.

13. *What role did leadership and top management commitment play in the process of the circular business model initiation?*

Choose an appropriate item:

14. *How was the new circular business model involved in the existing strategic plan of the company?*

Choose an appropriate item:

15. *What environmental drivers have inspired you for the circular business model development?*

Click or tap here to enter text.

16. *What was the biggest perceived risk(s) in the initiation stage?*

Click or tap here to enter text.

17. *Which biggest barrier(s) was perceived in the initiation stage?*

Click or tap here to enter text.

Ideation – relevant for persons who participated in this phase

Circular business model canvas building blocks:

1. *If you developed scenarios for the circular value offering, how did you assess the scenarios (the alternative plans)?*

Choose an appropriate item:

2. How difficult was it to define criteria for selecting the most promising customer segment(s) that your business model is aiming at primary?

Choose an appropriate item:

3. Which aspect of customer relations did you find most crucial for the business model success in the ideation phase?

☐ Trust ☐ speed of reaction ☐ honesty ☐ security ☐ privacy ☐ friendliness
☐ question of establishing new relations to customer segment ☐ level of automatization of customer relations
☐ level of personalization in customer relations ☐ cost of customer relationships
☐ question of how to integrate new customer relationship into the business model ☐ other, please, specify:.....

4. What limits did you set in the ideation phase for redesigning the supply chain and its structure?

Choose an appropriate item:

5. Did you consider new key partners for the new circular business model in the ideation phase?

Choose an appropriate item:

- 6 a) Which key activities were the most difficult to (re)design because of complexity (interference with many business model elements)?

☐ research and development ☐ production ☐ marketing ☐ sales & customer services
☐ other, please specify:

- 6 b) Which key activities were the most difficult to (re)design because of missing or inappropriate methods/tools?

☐ research and development ☐ production ☐ marketing ☐ sales & customer services
☐ other, please specify:

- 6 c) For which key activities (re)design was the lack of data most crucial?

☐ research and development ☐ production ☐ marketing ☐ sales & customer services
☐ other, please specify:

7. a) Which of your resources/ capabilities did you perceive as essential for the decision to undergo the new circular business model?

☐ research&development capabilities ☐ human resources ☐ partners relationship ☐ financial IT and ICT
☐ logistics and supply chain resources (tangible) ☐ logistics and supply chain capabilities
☐ marketing capabilities ☐ production capabilities ☐ service providing capabilities
☐ sourcing capabilities ☐ other...if yes, please, specify:.....

b) Which of your resources were seen as insufficient for the circular business model in the ideation stage?

☐ research&development capabilities ☐ human resources ☐ partners relationship ☐ financial IT and ICT ☐ logistics and supply chain resources (tangible) ☐ logistics and supply chain capabilities ☐ marketing capabilities ☐ production capabilities ☐ service providing capabilities ☐ sourcing capabilities ☐ other...if yes, please, specify:.....

8. To what extent was the profitability of the circular model evident in the ideation phase of model development?

Choose an appropriate item:

Was profitability quantitatively assessed in the early stages of model development?

☐ Yes ☐ No

9. a) Did you elaborate on revenue streams and in the ideation phase of the business model development?

☐ Yes ☐ No

b) Did the circular business model include a new (category of a) revenue stream for your company?

☐ Yes ☐ No

Here you can write your comments, ideas to the individual building blocks:

Click or tap here to enter text.

Additional questions:

10. How many problems there were with overcoming the existing business logic and with thinking out-of-the-box?

Choose an appropriate item:

11. How difficult was it to think in the complex and whole circular business model (and not solely and typically in new product development)?

Choose an appropriate item:

12. If yes, what were the biggest difficulties?

Click or tap here to enter text.

13. Which more or less systematic tools to develop new business model ideas were applied in the ideation stage if any? If not, why not?

Click or tap here to enter text.

Integration – relevant for persons who participated at this phase

There is no specific question for this phase concerning individual business model canvas building blocks

How much it was clear and became a part of plans what circular value proposition for customers and other stakeholders will be created, delivered and captured by whom (also consider the key partners), how, where, when and why?

Choose an appropriate item:

Here you can write your comments or ideas:

Click or tap here to enter text.

Additional questions:

1. *Have you formulated mitigation plans for high impact and high-likelihood risks?*

Choose an appropriate item:

2. *Have you prepared any performance indicators for the evidence of the project failure and success?*

Choose an appropriate item:

3. *Have there been any necessary organizational changes incorporated into the plans for the circular business model implementation?*

New position in the organizational structure?

☐ Yes ☐ No

New department in the organizational structure?

☐ Yes ☐ No

New tasks for some positions?

☐ Yes ☐ No

New outsourcing of some activity/process?

☐ Yes ☐ No

New insourcing of some activity/process?

☐ Yes ☐ No

4. *Have there been any necessary managerial issues incorporated into the plans for the circular business model implementation?*

Training and education

☐ Yes ☐ No

New methods of management (for planning, organizing, controlling, motivating, communication etc.)

☐ Yes ☐ No

Need for some changes of corporate culture?

☐ Yes ☐ No

5. *Have there been any risks incorporated into the plans for the circular business model implementation?*

☐ Yes ☐ No

6. *If yes, have there been any methods for the project risks assessment involved in the integration phase?*

☐ Yes ☐ No

Implementation – relevant for persons who participated at this phase

Circular business model canvas building blocks:

1. *What elements of the circular value proposition have to be added/adapted/changed during the implementation stage? Think about the environmental, economic and social elements.*

Environmental:

Click or tap here to enter text.

Economic

Click or tap here to enter text.

Social

Click or tap here to enter text.

2. *Is there any new, unexpected information about your customer segments are you discovering during the implementation stage?*

Choose an appropriate item:

3. *Are there any unexpected requirements/needs to build and maintain new ways for customer relationships (e.g. additional service – care, additional information etc.)- different from the plan?*

Choose an appropriate item:

4. *Are there any unexpected changes in channel structure, channel members' requirements/needs when delivering circular value to customers – different from the plan?*

Choose an appropriate item:

5. *Are there any unexpected changes in key partners' requirements/needs when creating, delivering and capturing circular value– different from the plan?*

Choose an appropriate item:

6. *Are there any new partners for circular value creation, value delivery and value capture – different from the plan?*

Choose an appropriate item:

7. *Are there any new, unexpected key activities for circular value creation, value delivery and value capture – different from the plan?*

Choose an appropriate item:

8. *Are there any new, unexpected, needed key resources for circular value creation, value delivery and value capture – different from the plan?*

Choose an appropriate item:

9. *Have been some changes in the cost structure needed during the implementation of the circular business model – not expected in the plan?*

Choose an appropriate item:

10. *Have some new revenue streams or some changes in the revenue streams occurred during the implementation stage? (speed, amount, character etc.)*

Choose an appropriate item:

Here you can write your comments, ideas to the individual building blocks:

Click or tap here to enter text.

Additional questions:

11. Have there been any necessary organizational changes incorporated into the circular business model implementation – different from the plan?

New position in the organizational structure?

☐ Yes ☐ No

New department in the organizational structure?

☐ Yes ☐ No

New tasks for some positions (e.g. marketing, production, logistics, sourcing etc.)?

☐ Yes ☐ No

New outsourcing of some activity/process?

☐ Yes ☐ No

New insourcing of some activity/process?

☐ Yes ☐ No

Otherwhat specifically?.....

☐ Yes ☐ No

12. Have there been any necessary managerial issues incorporated into the circular business model implementation - different from the plan?

Training and education

☐ Yes ☐ No

New methods of management (for planning, organizing, control and controlling, motivating, communication etc.)

☐ Yes ☐ No

Need for some changes of corporate culture?

☐ Yes ☐ No

Otherwhat specifically?.....

☐ Yes ☐ No

13. Are there any new risks with the implementation of the circular business model –different from the plan?

☐ Yes ☐ No

Changes

What are the:

1. *biggest changes during the implementation phase when compared to the initial assumptions and plans (do not mention covid-19)?*

Click or tap here to enter text.

2. *biggest challenges/barriers during the implementation phase when compared to the initial assumptions and plans (do not mention covid-19)? How were these challenges solved, and with what effects?*

Click or tap here to enter text.

In both cases, please, describe them individually, if possible:

- j) Circular value proposition
- k) Customer segment
- l) Customer relationships
- m) Channel
- n) Key partners
- o) Key activities
- p) Key resources
- q) Cost structure
- r) Revenue streams

3. *Do you consider the circular business model development process as different to other business model innovation processes in the company?*

Choose an appropriate item:

If yes, what was different and why?

Click or tap here to enter text.

4. *How did you overcome internal but also external (key partners) resistance during the whole process – from the initiation till the implementation?*

Choose an appropriate item:

If there was some internal or external resistance, what was the problem?

Click or tap here to enter text.

5. *Are there any challenges related to the simultaneous management of the existing business model and the new model? Any conflict of interests between the two business models, for example?*

☐ Yes ☐ No

If yes, what challenges and or conflict of interests?

Click or tap here to enter text.

6. *Are there any new beneficiaries – new recipients – stakeholders of the circular value your business model generates?*

☐ Yes ☐ No

If yes, who and what value they obtain?

Click or tap here to enter text.

Risks and impact of change

1. What are the:
biggest risks during the implementation phase when compared to the initial assumptions and plans (do not mention covid-19)?

Click or tap here to enter text.

2. *Has an impact value chain (input – activity – output – outcome – impact) been created?*

☐ Yes ☐ No

If not, why not?

Click or tap here to enter text.

3. *How do you assess the impacts of the transformation process? Ex-ante or ex-post or during?*

☐ Ex-ante ☐ ex-post ☐ during

4. *Do you use any proxy indicators to assess the impact in monetary terms?*

☐ Yes ☐ No

If so, which ones and for which activities?

Click or tap here to enter text.

Appendix D – Business model risks

BM Areas	Groups of business model risk and uncertainty factors	
	Groups of BM risk and uncertainty factors	Description
Customer	Customer demand risk and uncertainty factors	All factors related to misinterpreting or not meeting customer demand, such as not focusing on the customer and customer needs, not attracting or retaining customers, or not meeting the community requirements
	Customer relationship risk and uncertainty factors	All factors that can damage the relationship with customers, such as losing the customer interface, opportunistic or adverse behavior on the side of the customer, or inflexible agreements
	Customer solvency risk factors	All risk factors related to the customer solvency, including bad credit rating, insolvency, or default of the customer
	Customer access risk factors	All risk factors that hinder the customer's access, such as missing market access, a strong intermediary, high market entry barriers, or a strong competitor
	Multiple BM risk factors	Critical aspects attributed to the emergence of multiple BMs within one organization, causing cannibalization of the existing customer base, losing loyal customers, or offending existing customers by becoming a competitor
Offer	Quality risk and uncertainty factors	All aspects that are related to the quality of an offer, such as a gap in expected vs. delivered performance, durability, and functionality
	Availability and maintenance risk factors	All critical aspects that relate to the availability and maintenance of an offer as an important component of the value proposition and the related consequences of the offering's poor performance
	Data risk and uncertainty factors	All risk and uncertainty factors related to data usage such as data security, data ownership, data privacy, and data quality
	(New) technology risk and uncertainty factors	All risk and uncertainty factors that are linked to the utilization of technologies that are new or still in a premature state, highly complex, or for which the company lacks experience
	Innovativeness risk factors	All risk factors linked to the dependency on the innovativeness of the value proposition of a company or BM (e.g., not using state-of-the-art technology)
Infrastructure	BM ecosystem risk and uncertainty factors	All risk and uncertainty factors within a business ecosystem and value network that constitute multilateral relationships between the different parties
	Bilateral cooperation risk and uncertainty factors	All critical aspects within the cooperation and bilateral relationship with partners or customers that impact the stability of the relationship
	Capabilities and resources risk and uncertainty factors	Factors related to critical capabilities or resources required to realize a certain BM, such as special equipment, data, or human and technical resources
	Operational risk and uncertainty factors	Factors related to the operations in business processes that are often connected to errors in human or technical behavior

Table 1 (continued)

BM Areas	Groups of business model risk and uncertainty factors	
	Groups of BM risk and uncertainty factors	Description
	Intellectual property (IP) risk and uncertainty factors	Concerned with the drainage of IP or know-how from the BM managers' point of view; Such IP and knowledge also includes sensitive data on the firm's customers and partners
	BM portfolio risk factors	All risk factors that are linked to the management of a wide variety of offerings and the complexity and defocusing related to a BM portfolio
Financial Viability	Financial resources risk factors	These factors comprise all aspects that impact the BM manager's capability to finance the BM
	Investment risk and uncertainty factors	Factors linked to high investments and capital tie-up necessary to create, deliver, or capture a value proposition of the BM, including production plants or inventory costs
	Monetization risk and uncertainty factors	Include all factors that could risk the monetizability of a certain value proposition or offer
	Revenue mechanism risk factors	Factors related to the revenue mechanism of a BM and its related taxation regulations, which influences the buying sensitivity for a certain value proposition and the choice of a revenue model and has an impact on taxation and accounting as there may be model-specific regulations a company has to comply with
	Pricing risk and uncertainty factors	This category comprises all risk and uncertainty factors related to the price setting of a BM value proposition and directly corresponds to the customer price sensitivity and its value perception of the value proposition of the BM.
	Lifecycle risk and uncertainty factors	This group describes lifecycle risk and uncertainty factors that come with a possible uncalculated change of costs and revenues in the lifecycle of a BM.
Environment	Political risk and uncertainty factors	Factors that include aspects related to political, social, or governmental factors
	Environmental risk and uncertainty factors	Factors that cover all aspects linked to acts of nature
	Economic risk and uncertainty factors	This group includes all risk and uncertainty factors in the business environment or market (e.g., customs, currency exchange rates, taxation, economic cycle)
	Legal and regulatory risk and uncertainty factors	All aspects regarding the commitment of regulatory and legal circumstances
	Competition risk and uncertainty factors	Factors that cover all critical aspects relating to existing or new competitors
	Technological change risk and uncertainty factors	This group covers all aspects related to technological change

Source: Based on Osterwalder and Pigneur (2010)

Source: Brillinger et al, 2020